



Lecture 11:

Controlling Image Generation

Email: idadschwartz@gmail.com or
idan.schwartz@biu.ac.il

Recall: conditioning with diffusion models

- **Classifier-free** : Adds conditioning; requires supervised data

$$\nabla \log p(\boldsymbol{x}_t \mid y) = \underbrace{\gamma \nabla \log p(\boldsymbol{x}_t \mid y)}_{\text{conditional score}} + \underbrace{(1 - \gamma) \nabla \log p(\boldsymbol{x}_t)}_{\text{unconditional score}}$$

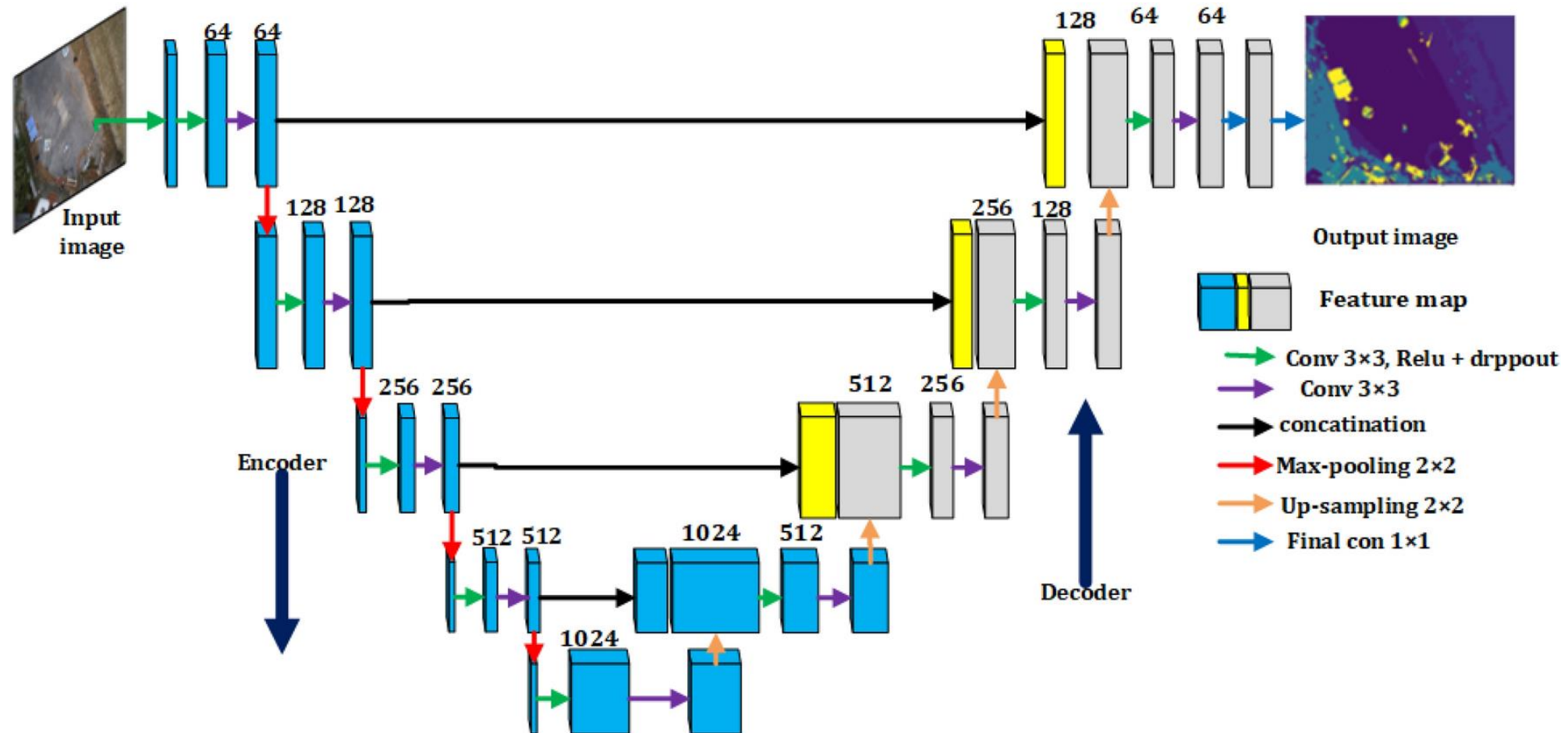
A stained glass window of a panda eating bamboo.



$\gamma = 1$

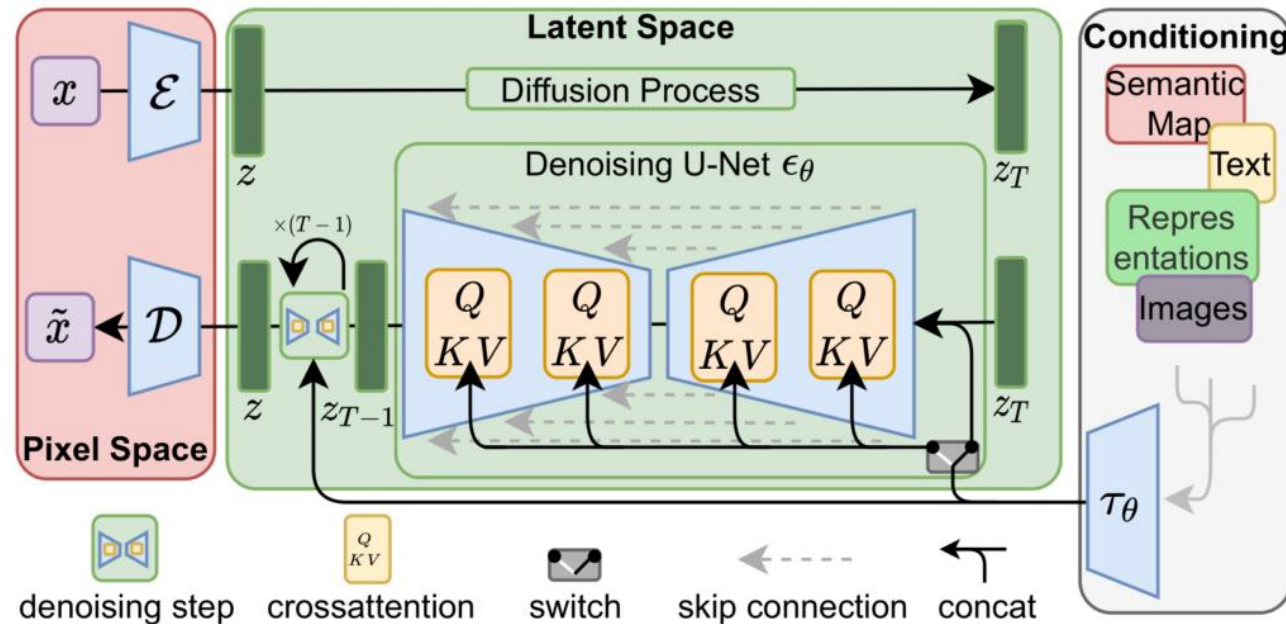
$\gamma > 1$

UNet: The architecture that predicts the noise, often used for segmentation



Recall: Latent Diffusion Model (Stable Diffusion)

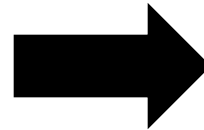
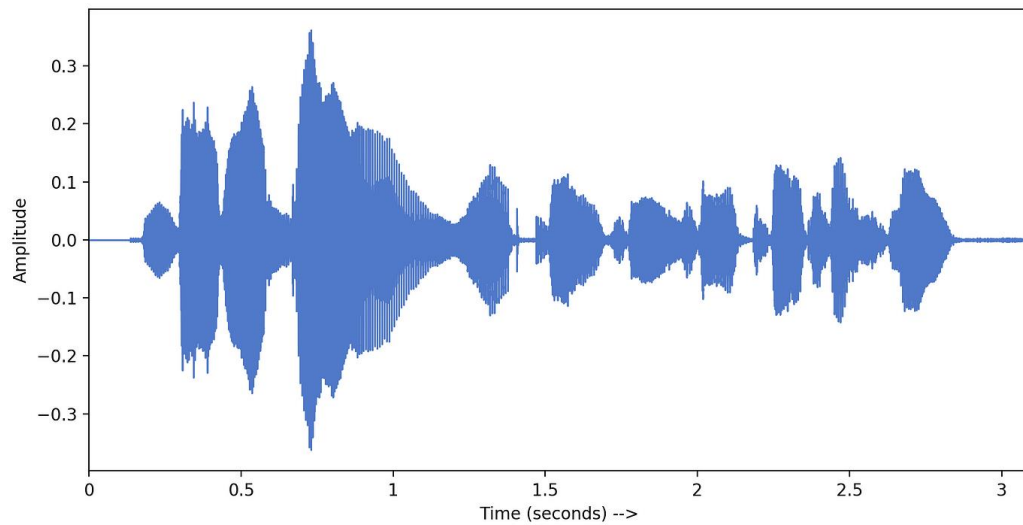
- Latent down sample images by 8x spatially (256 -> 32, 512 -> 64)
- UNet architecture with Transformers
- Text conditioning is done with cross-attention
- Text features using CLIP text encoder



Improve controllability in diffusion models

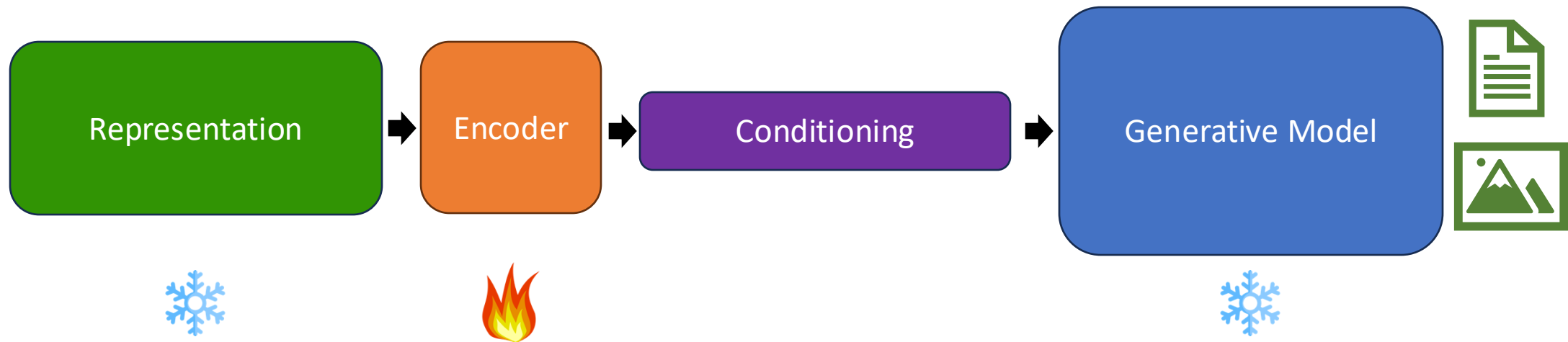
- Add modalities
- Use additional conditions to generate alongside prompts
 - Image-to-image synthesis
 - Auxiliary conditions (pose, depth-map, etc.)
- Personalization (e.g., people pets, items)
- Respect the prompt

Audio-to-image generation



How to enable multimodal conditioning?

- Train an encoder
 - **Cons:** Requires a fine-tuning step, supervised data
 - **Pros:** Immediate inference
- Can be useful as a **simple strong baseline**



Audio Representation

- We can use transfer learning from an audio classifier
- AudioSet consists of 632 audio classes

AudioSet

Human sounds

- Human voice
- Whistling
- Respiratory sounds
- Human locomotion
- Digestive
- Hands
- Heart sounds, heartbeat
- Otoacoustic emission
- Human group actions

Source-ambiguous sounds

- Generic impact sounds
- Surface contact
- Deformable shell
- Onomatopoeia
- Silence
- Other sourceless

Animal

- Domestic animals, pets
- Livestock, farm animals, working animals
- Wild animals

Sounds of things

- Vehicle
- Engine
- Domestic sounds, home sounds
- Bell
- Alarm
- Mechanisms
- Tools
- Explosion
- Wood
- Glass
- Liquid
- Miscellaneous sources
- Specific impact sounds

Music

- Musical instrument
- Music genre
- Musical concepts
- Music role
- Music mood

Natural sounds

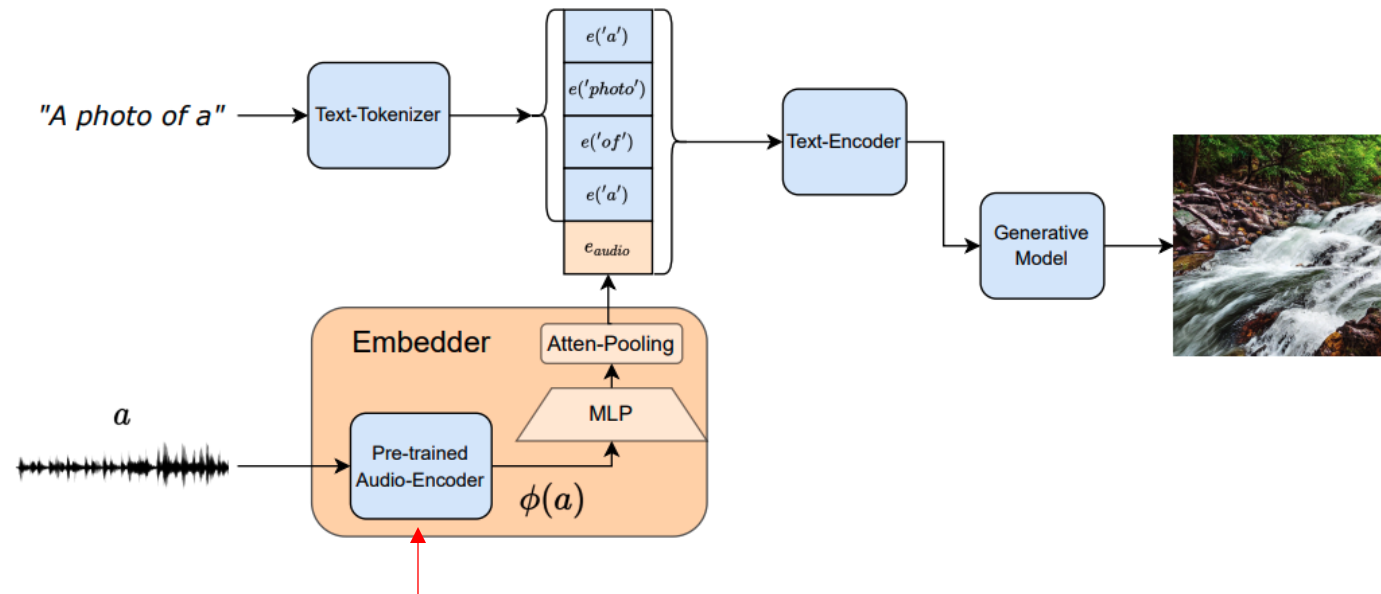
- Wind
- Thunderstorm
- Water
- Fire

Channel, environment and background

- Acoustic environment
- Noise
- Sound reproduction

Mapping Audio to a Token

- We train a mapper from audio to a token based on a generative loss
- Only the Embedder parameters are trained in the process



Representation from an AudioSet classifier

Results

Fine-grained audio details
E.g., type of guitar



Multiple entities
E.g., a girl and acoustic guitar



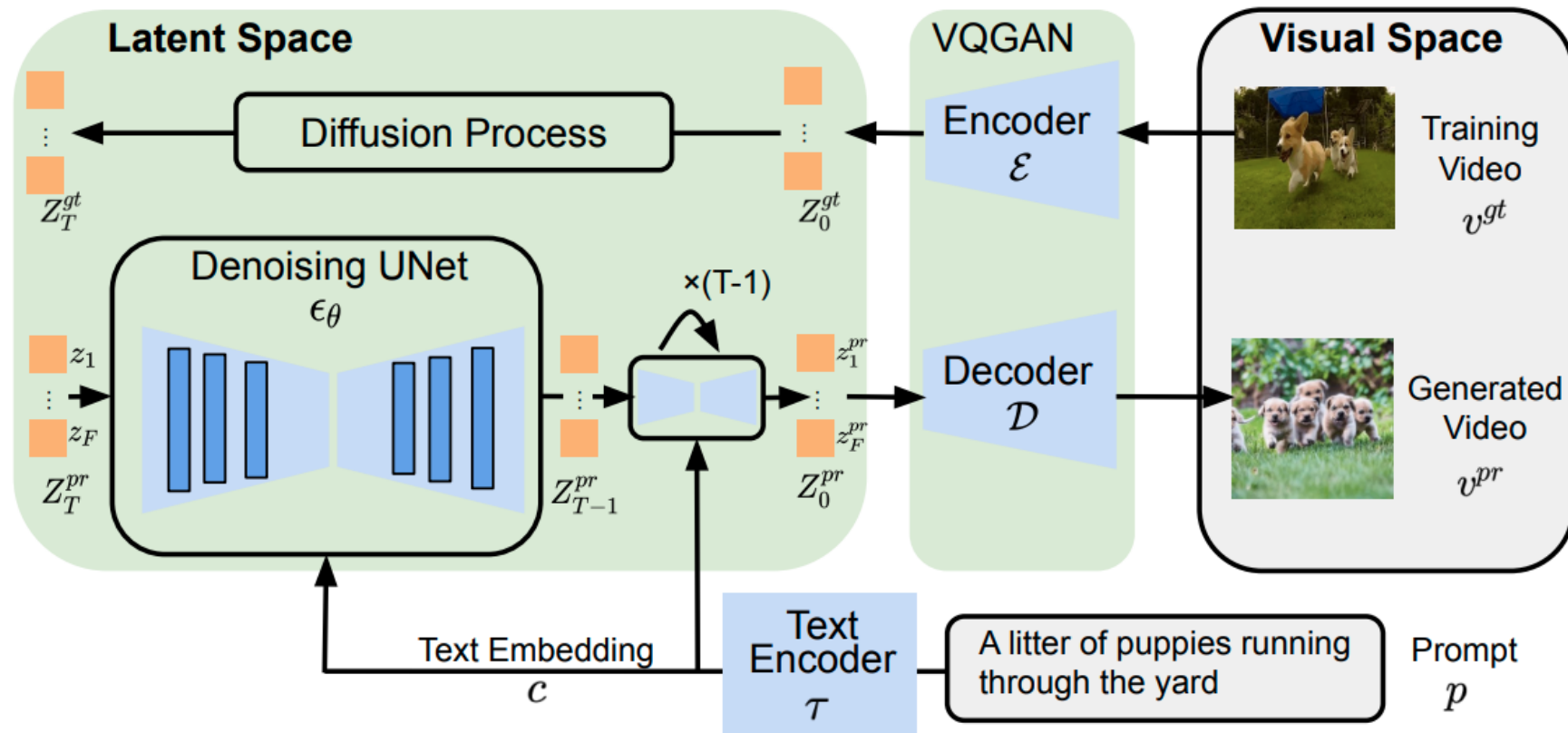
Objective Evaluation Functions

- We propose:

1. Audio-Image Semantic (AIC): compare classes
2. Audio-Image Similarity (AIS): compare representations
3. Image-Image Similarity (IIS): CLIP distance
4. Fréchet Inception Distance (FID): Compares the distribution of generated images with a set of real images ("ground truth").

Method	Metric			
	AIC ↑	FID ↓	AIS ↑	IIS ↑
Reference	54.66	-	-	-
SD (Text)	71.28	52.85	-	-
Wav2Clip	29.32	99.89	47.76	51.11
ImageBind	39.15	67.42	67.48	75.50
AUDIOTOKEN	45.48	56.65	68.23	76.66

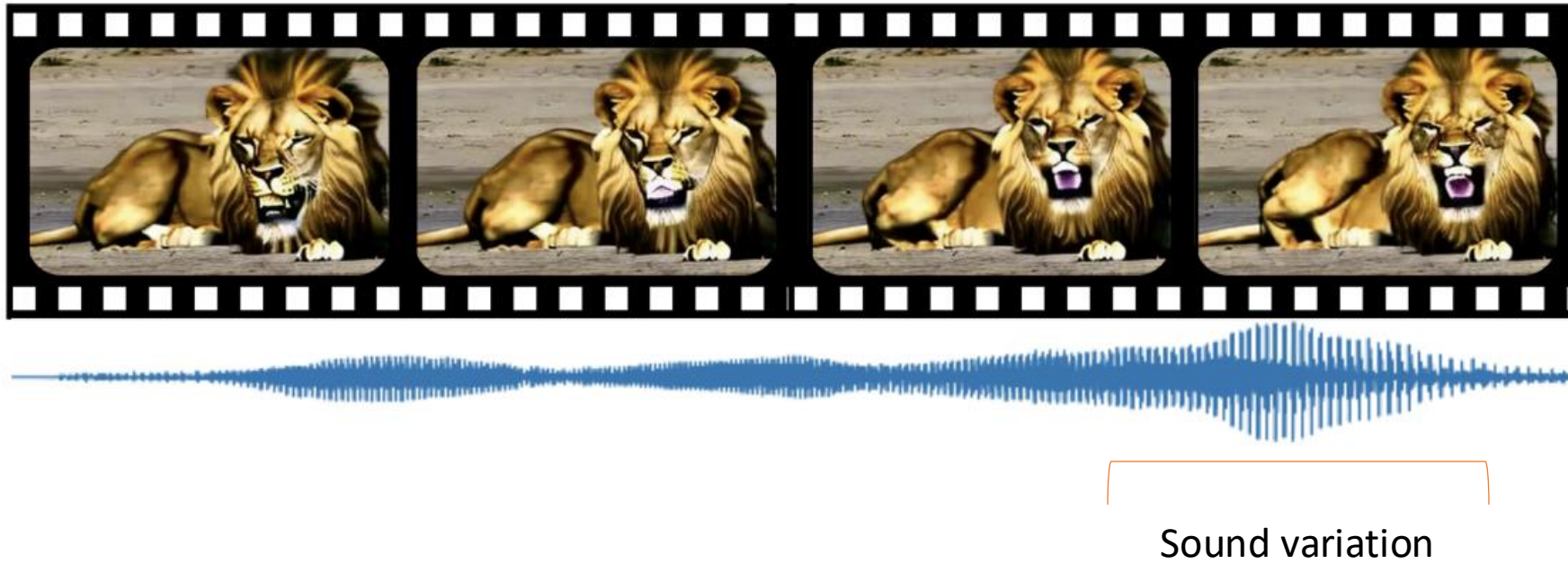
Generating videos with ModelScope*



* - ModelScope Text-to-Video Technical Report (Wang et al. 2023)

Audio-to-Video

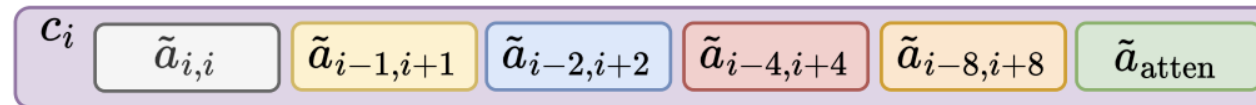
- Audio to video require temporal alignment



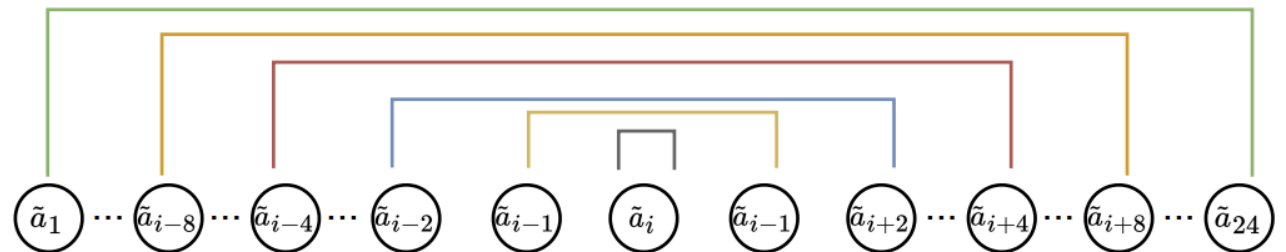
Temporal tokens



Generator

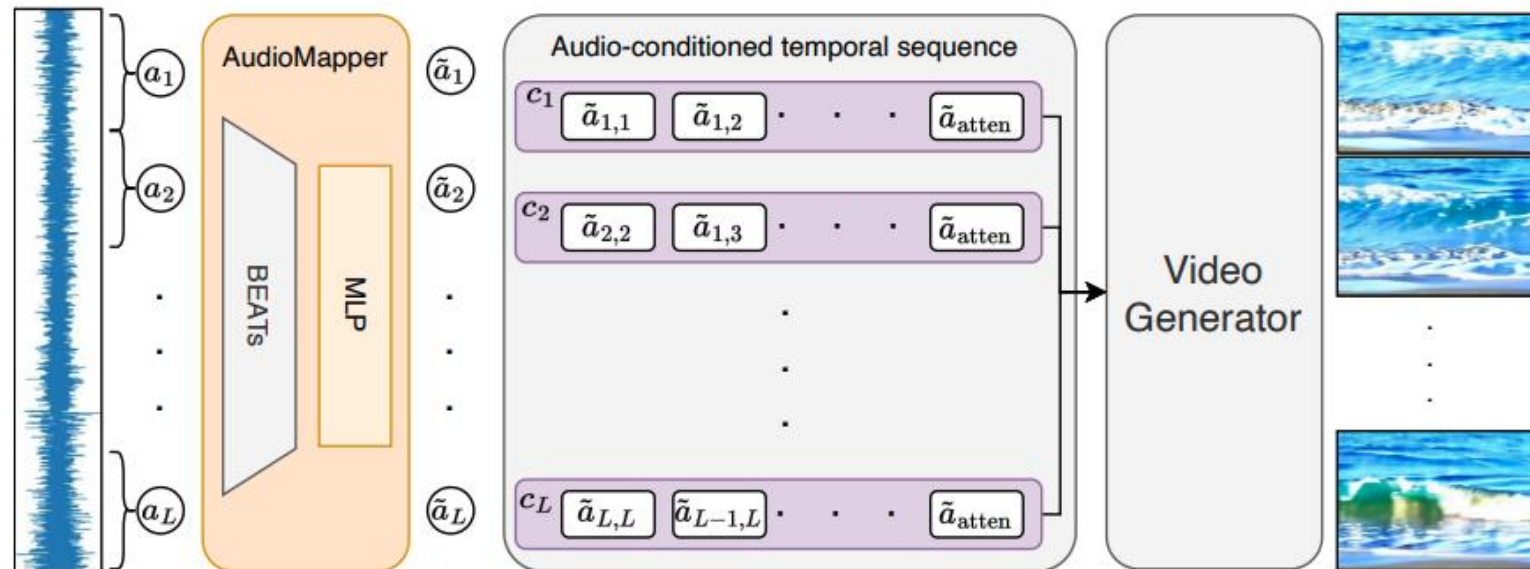


Context windows:



Audio-to-Video Architecture

- We repurpose a text-to-video model
- Temporal audio-tokens capturing a different audio segment



Audio-aligned generated Video

- We train the mapper on different datasets:

VGGSound



Landscape



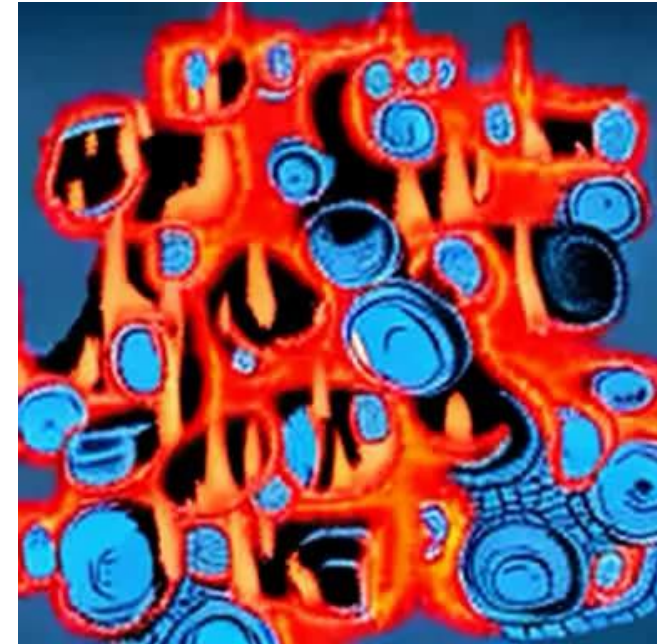
AudioSet-Drum



Multi-modal conditioning



A video of {  } on the moon

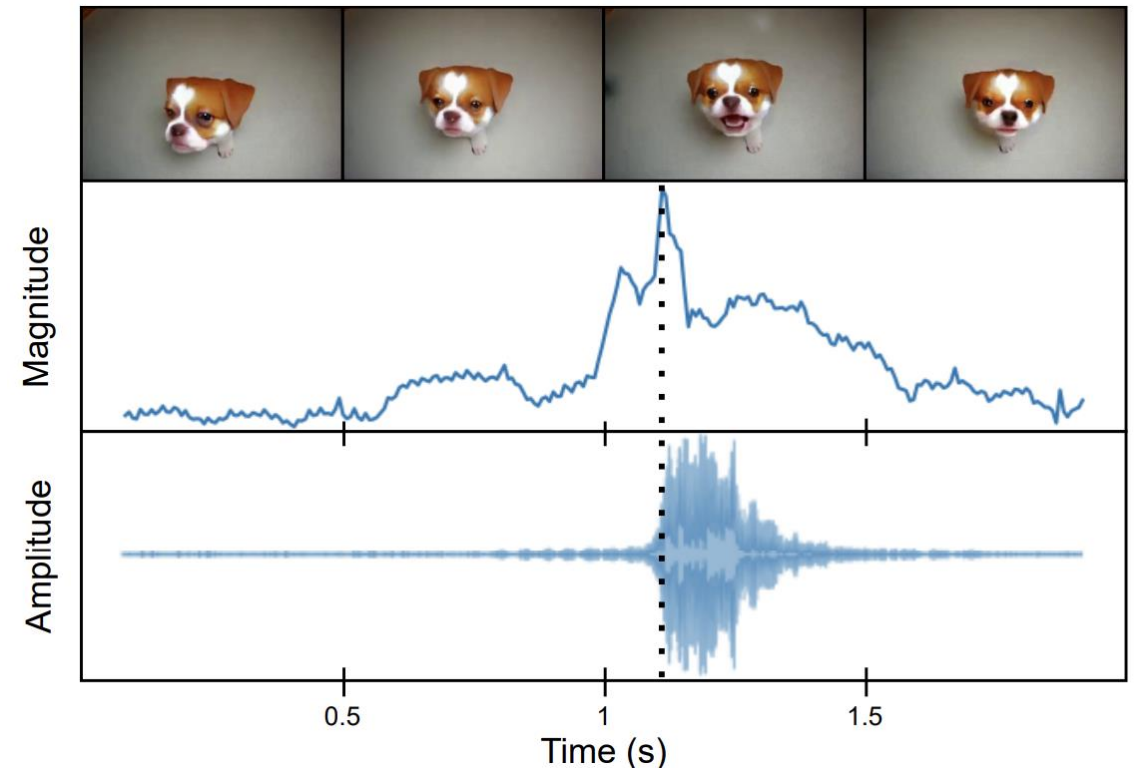


A painting of {  }

AV-Align

- Audio peaks $\mathcal{A}$ detected using an Onset Detection algorithm
- The Optical Flow magnitude mean is calculated for each video frame. The video peaks $\mathcal{V}$ identify rapid changes over time
- The alignment score is defined as:

$$\text{AV-Align} = \frac{1}{2|\mathcal{A} \cup \mathcal{V}|} \left(\sum_{a \in \mathcal{A}} \mathbf{1}[a \in \mathcal{V}] + \sum_{v \in \mathcal{V}} \mathbf{1}[v \in \mathcal{A}] \right)$$



Quantative Evaluation

1. Video quality and diversity
2. Audio-video semantic alignment
3. Audio-video temporal alignment

Model	FVD (↓)	CLIPSIM (↑)	IS (↑)	AV-Align (↑)
VGGSound				
ModelScope Text2Vid	801	0.69	15.55	0.27
ModelScope Random	1023	0.47	6.32	0.26
Ours	923	0.57	11.04	0.35
AudioSet-Drums				
TATS	303	0.69	2.10	0.28
Ours	299	0.70	2.78	0.61
Landscape				
MM-Diffusion	922	0.53	2.85	0.41
Ours	784	0.57	4.49	0.54

Going beyond text conditioning

What if we wanted to condition the generation process on a pose image along with language supervision?



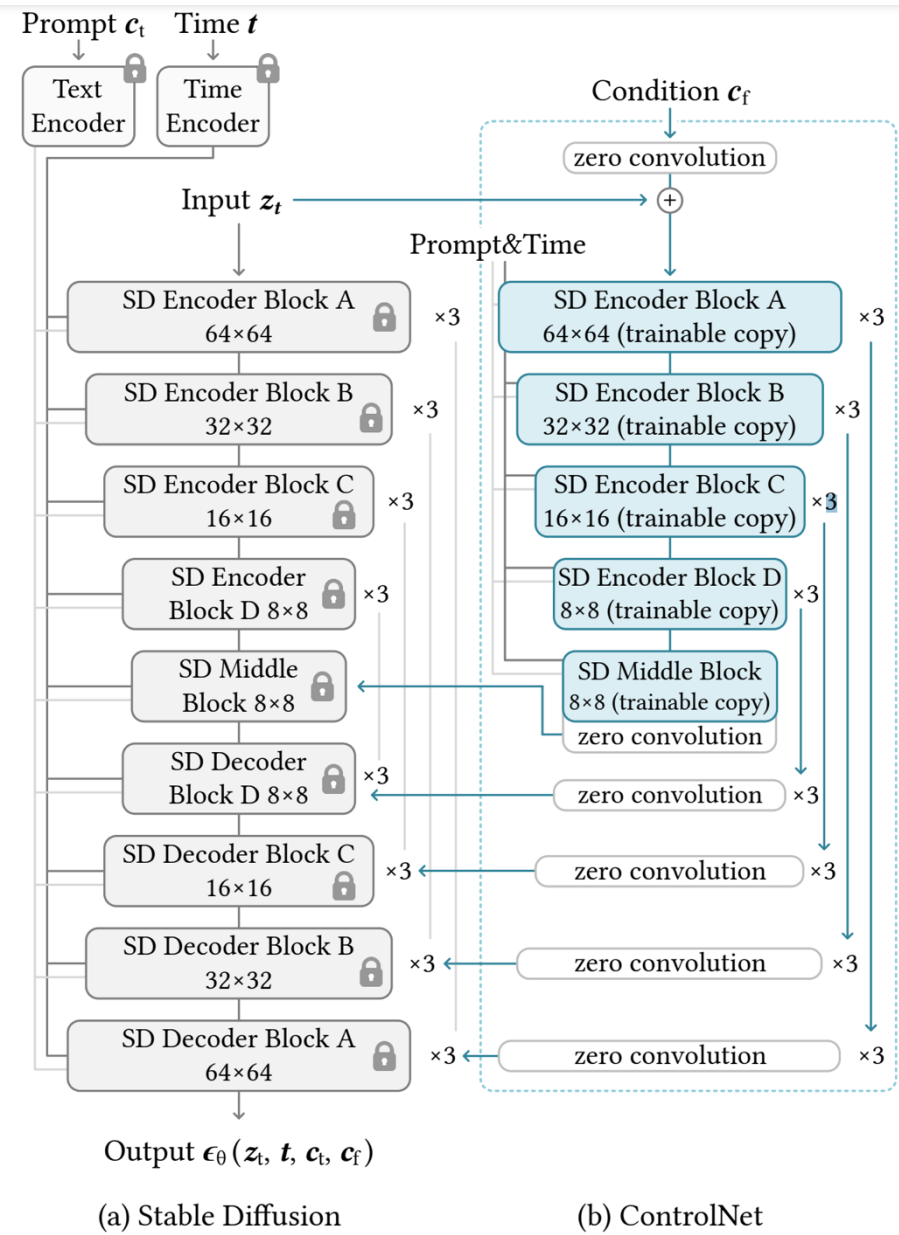
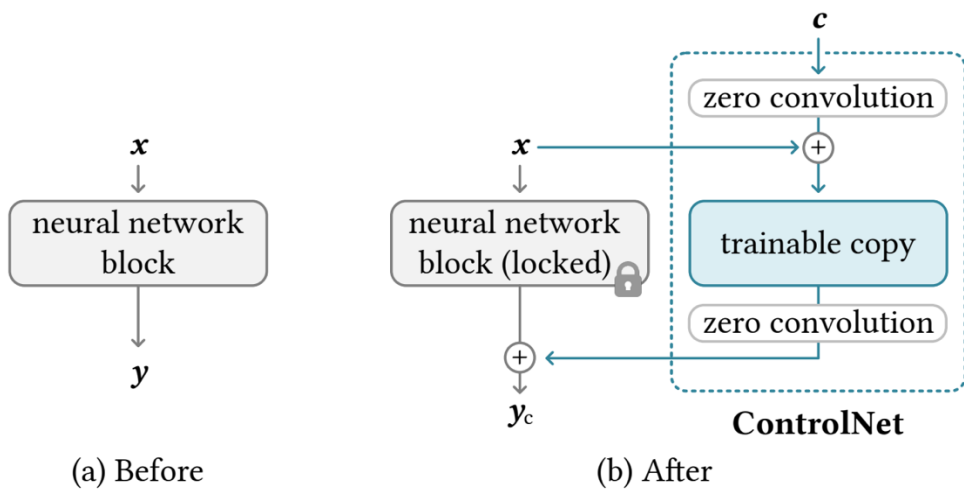
"Darth Vader in a desert"

ControlNet

Finetune a diffusion model to include extra conditioning:

- Canny edge map
- Depth
- Human Pose
- Lower Resolution Image
- Segmentation
- Sketch

ControlNet



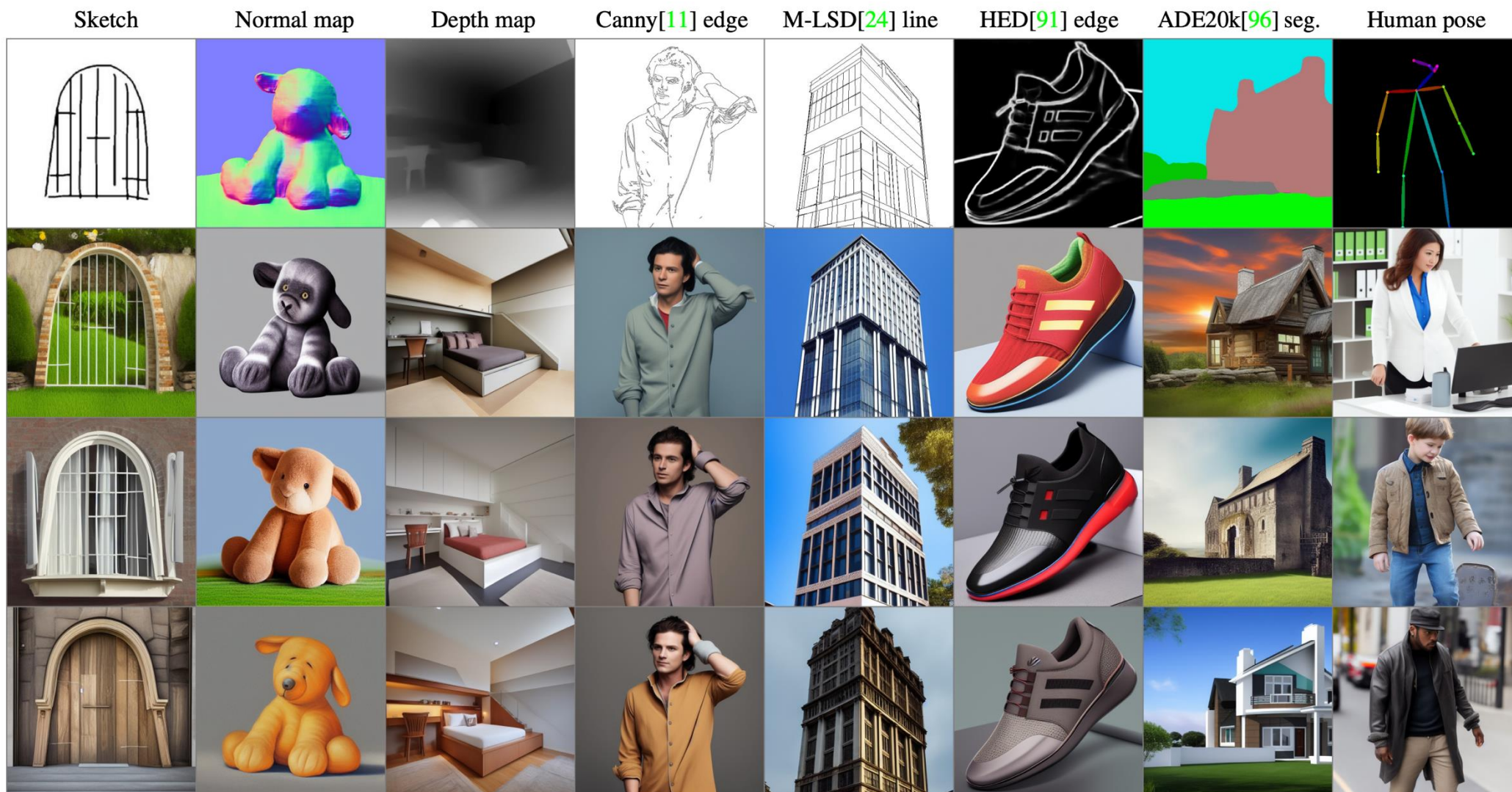
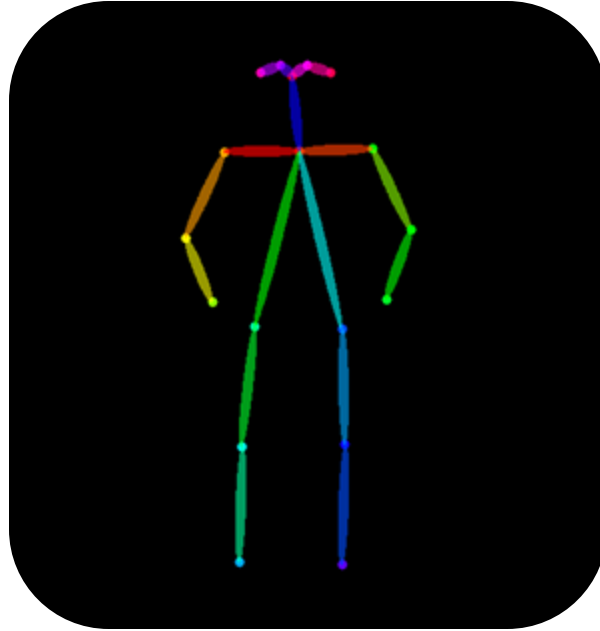


Figure 7: Controlling Stable Diffusion with various conditions **without prompts**. The top row is input conditions, while all other rows are outputs. We use the empty string as input prompts. All models are trained with general-domain data. The model has to recognize semantic contents in the input condition images to generate images.

Multiple controls



Canny map



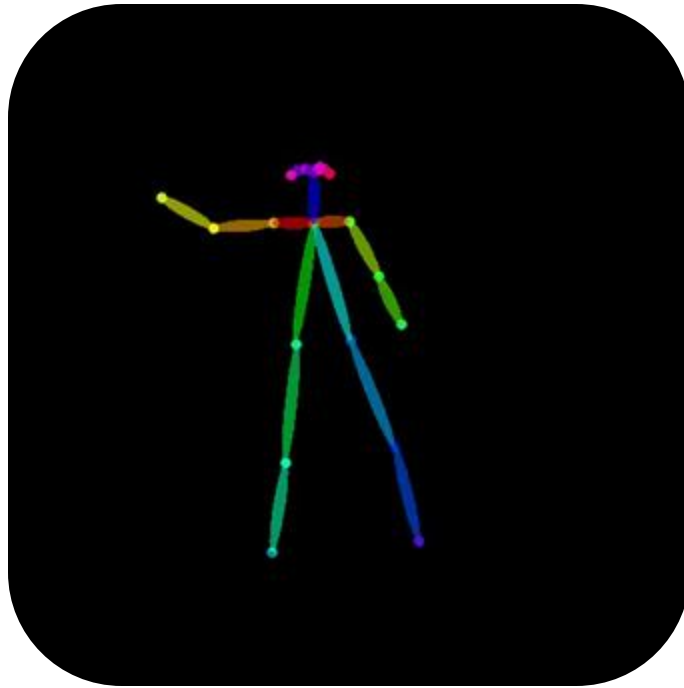
Pose



Final Image

"a giant standing in a fantasy landscape, best quality"

Could be used to make videos



Text2Video-Zero; Khachatryan et al.

Image conditioning

- Image encoder
 - Adapt image representations with a small “projection” module
- Decouple cross-attention
 - For each frozen cross-attention layer, add a *new* cross-attention layer
- Training
 - Only train the projection module and the new cross-attention layers (~22M)

IP-Adapter

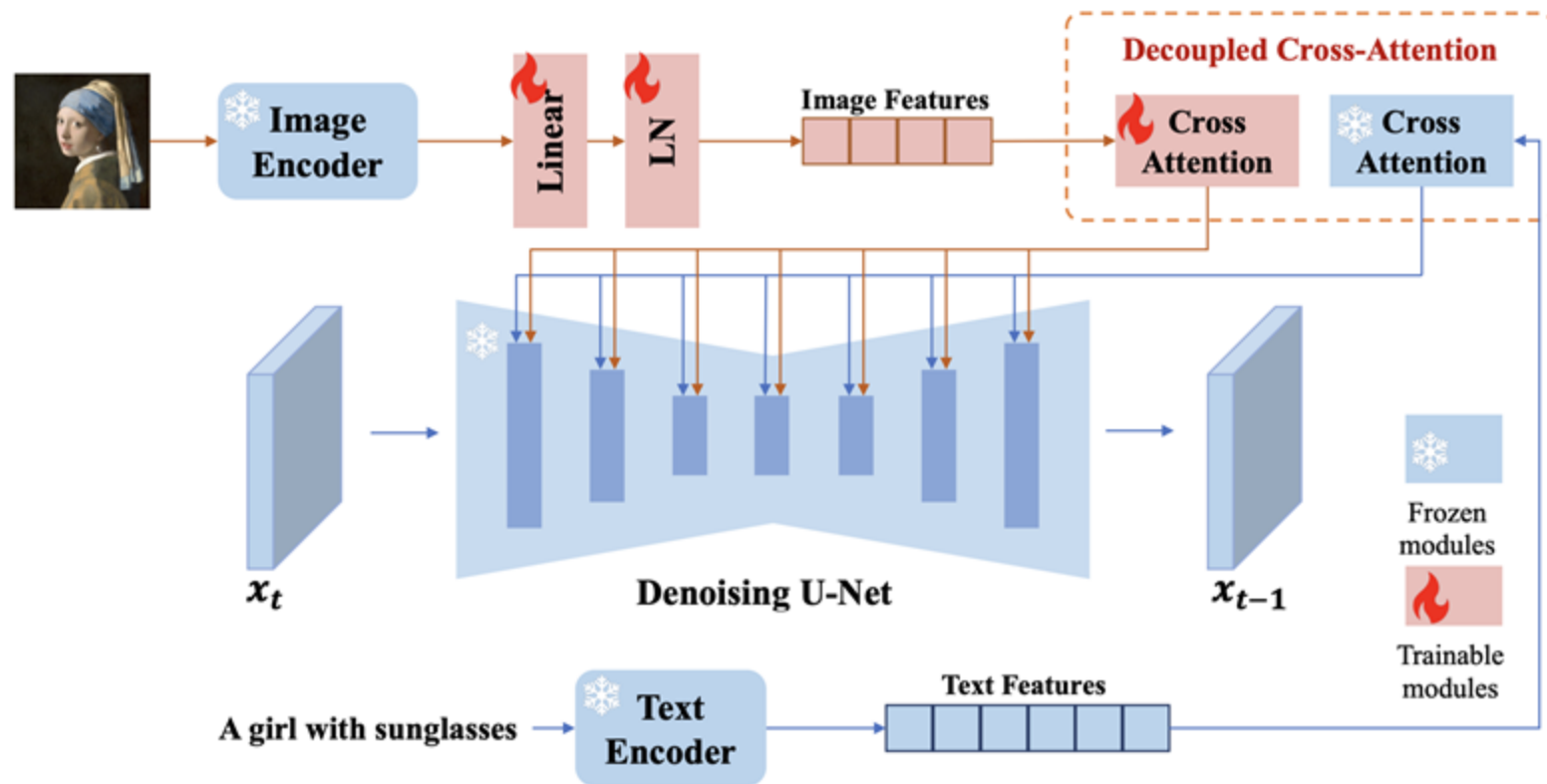


Image prompting

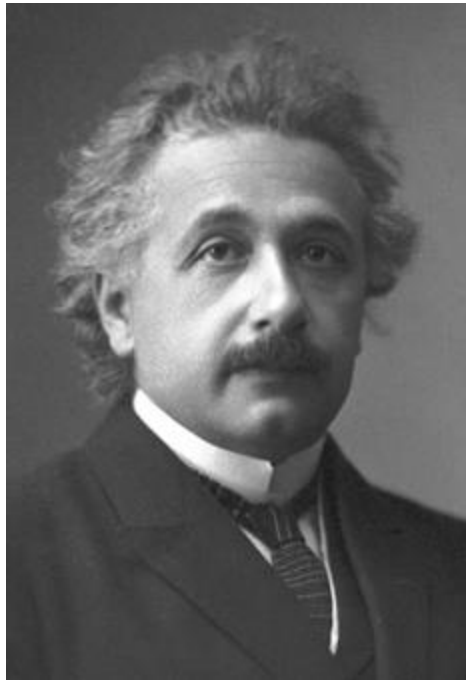
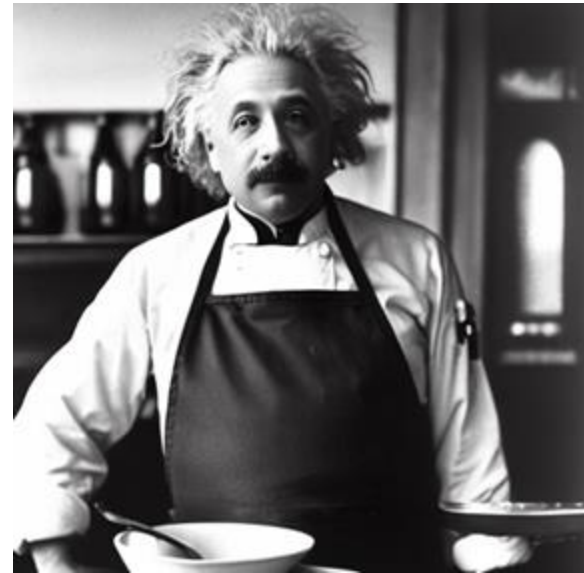


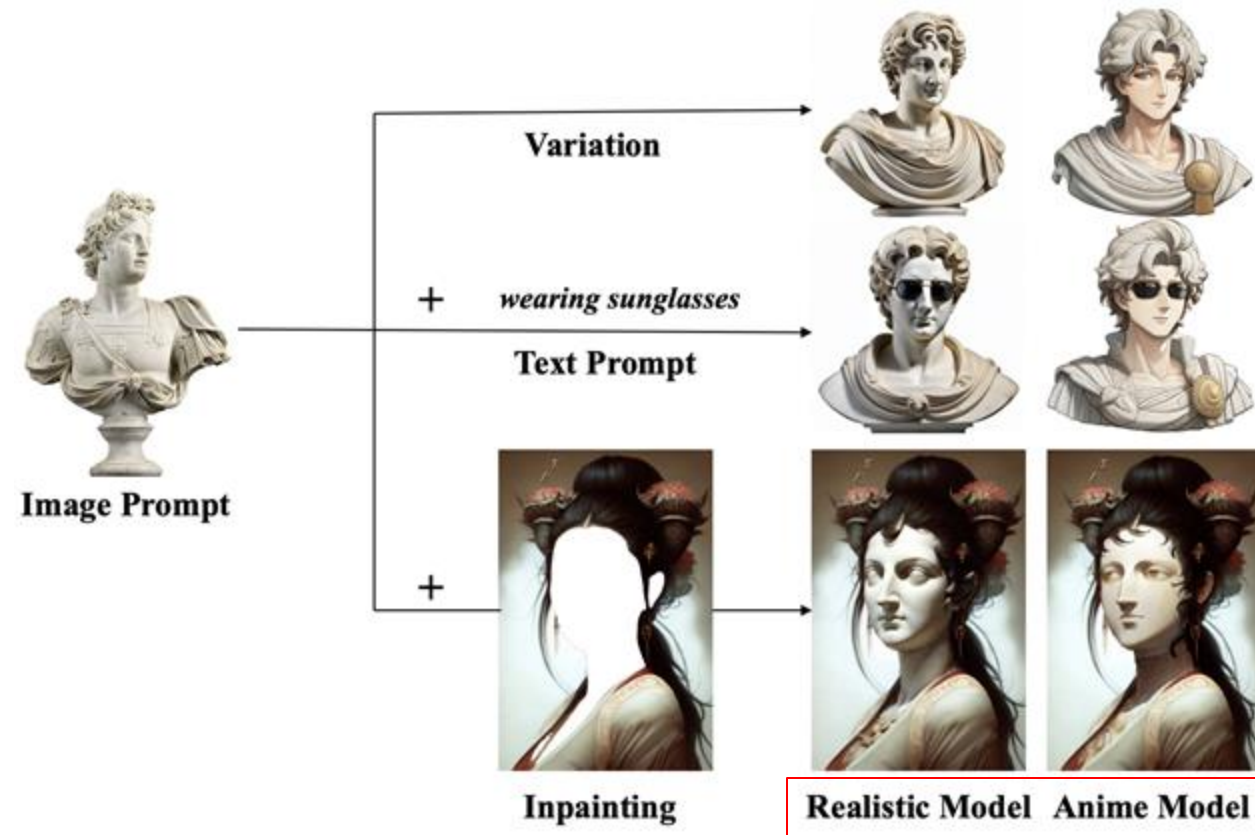
Image prompt



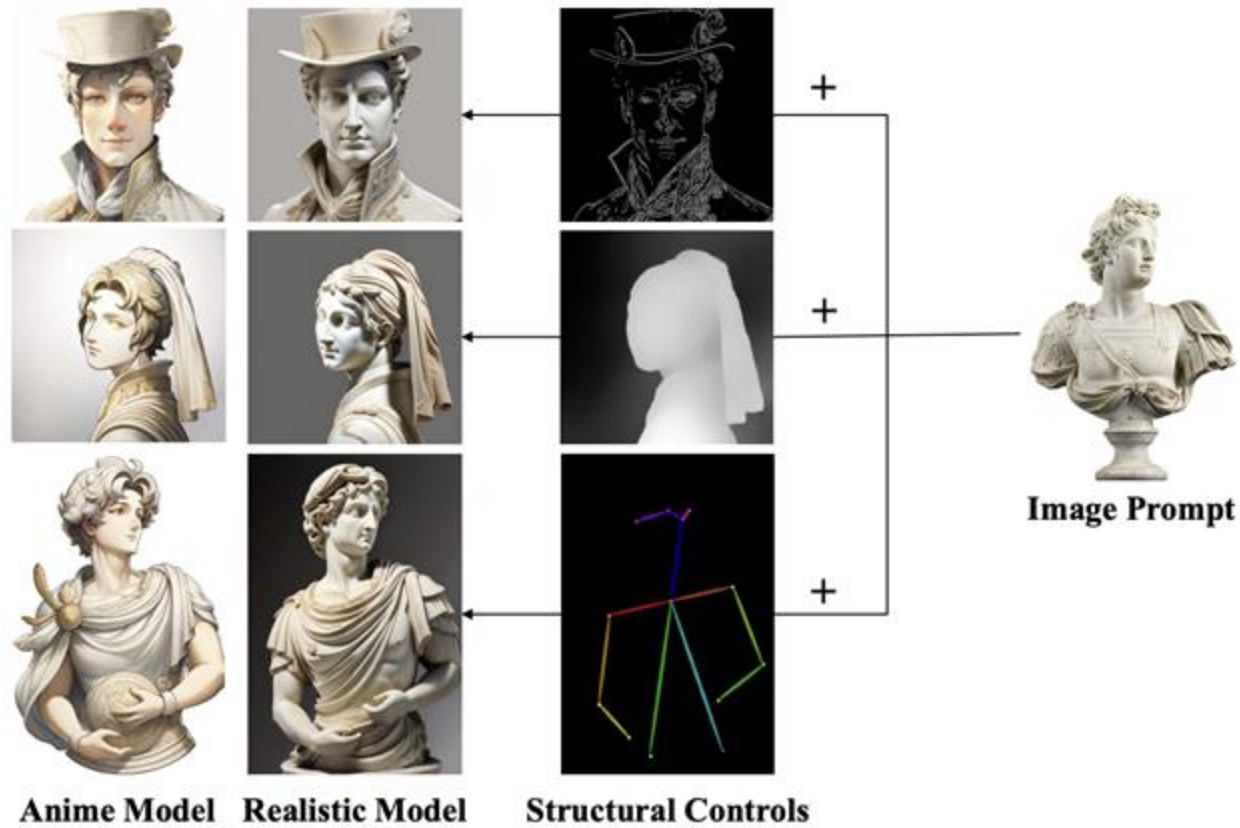
Output

“A photo of Einstein as a chef, wearing an apron, cooking in a French restaurant”

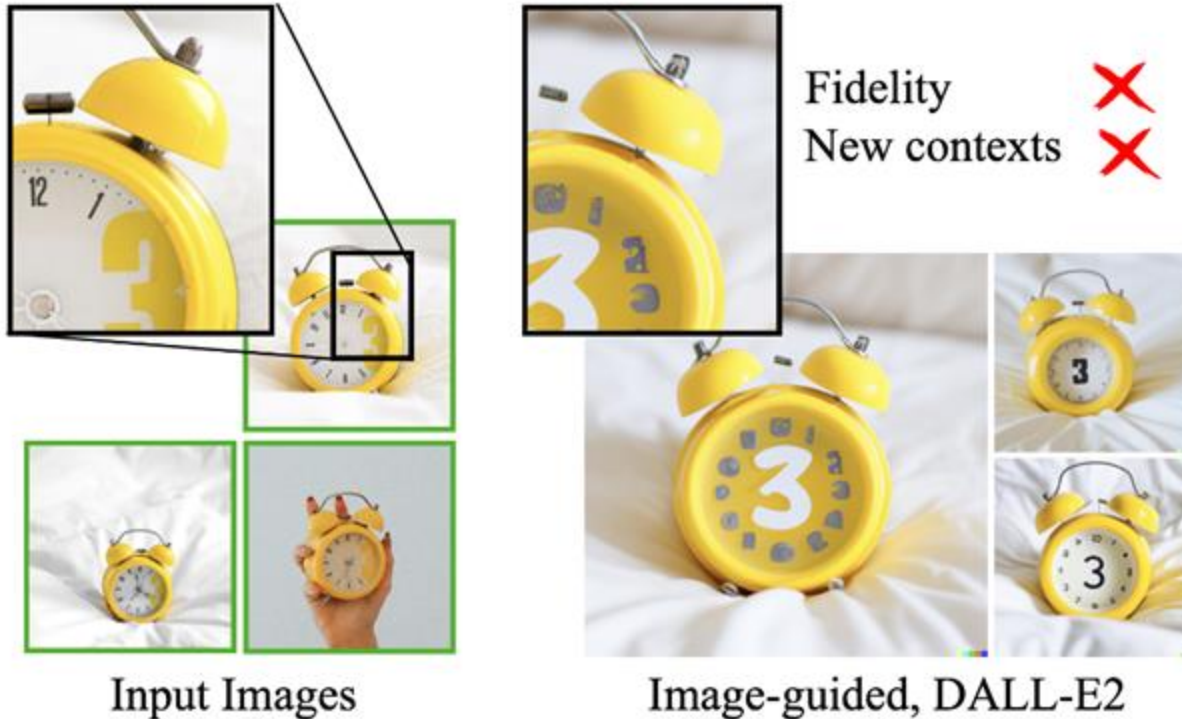
Image prompting with fine-tuned models



Combine with additional adapters like ControlNet



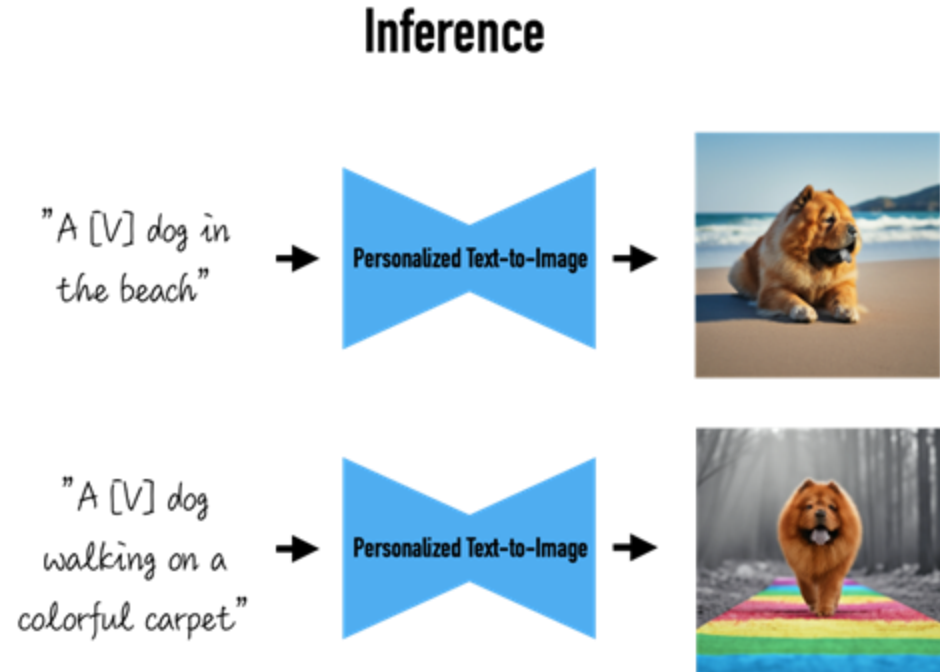
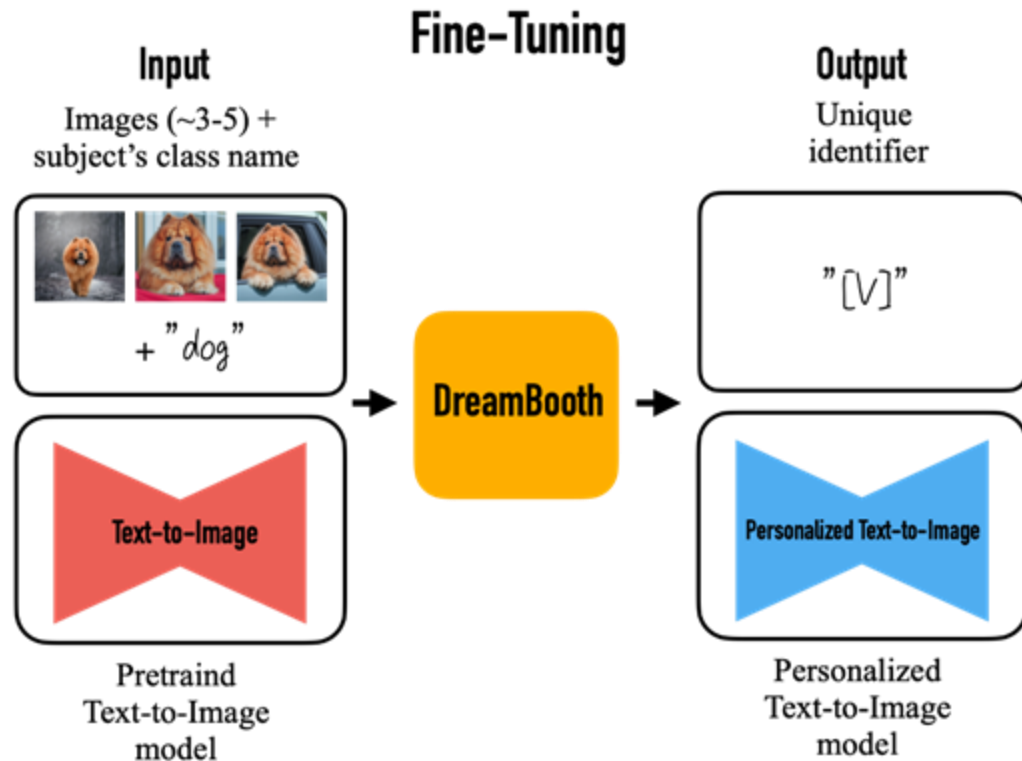
Concept learning: Subject-driven generation for personalization



- Learn new concepts
- Introduce *personalization*.

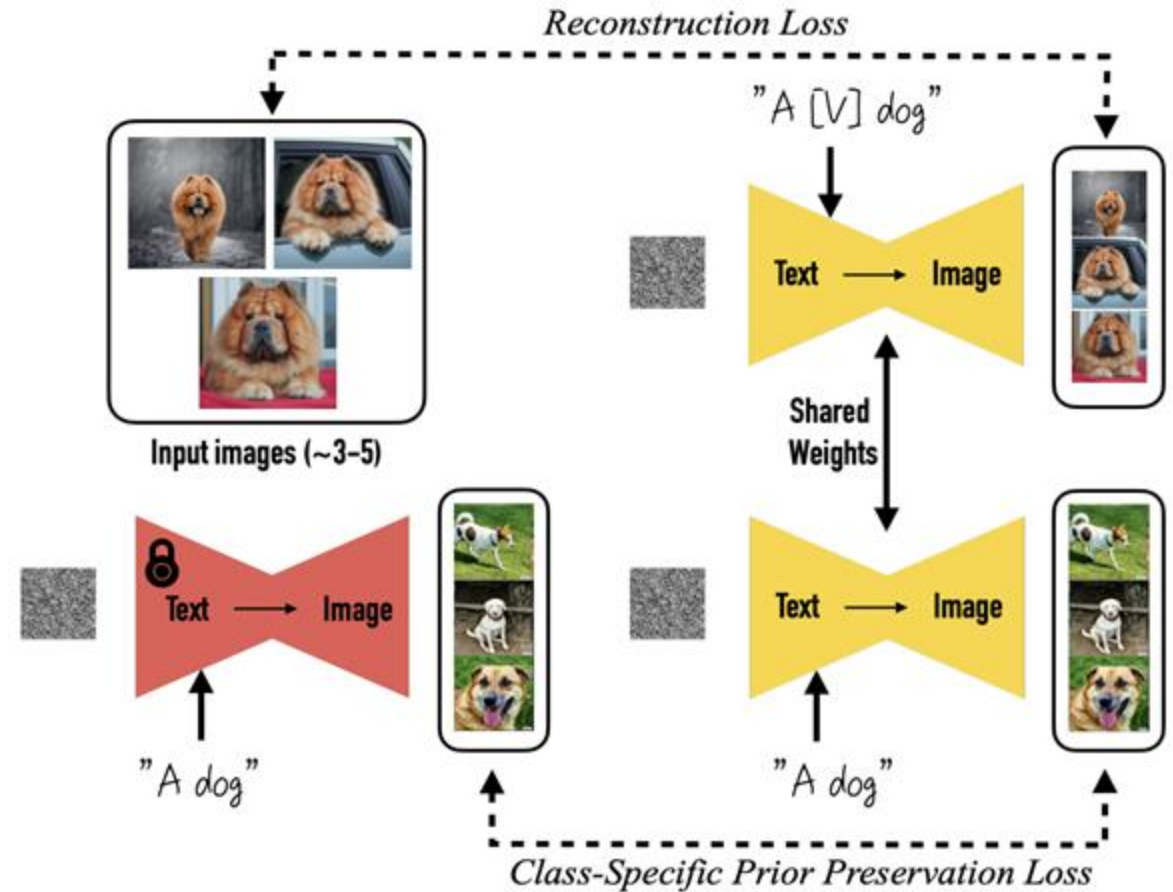
Subject-driven generation for personalization

- Learn a new concept from a small 3-5 set of image



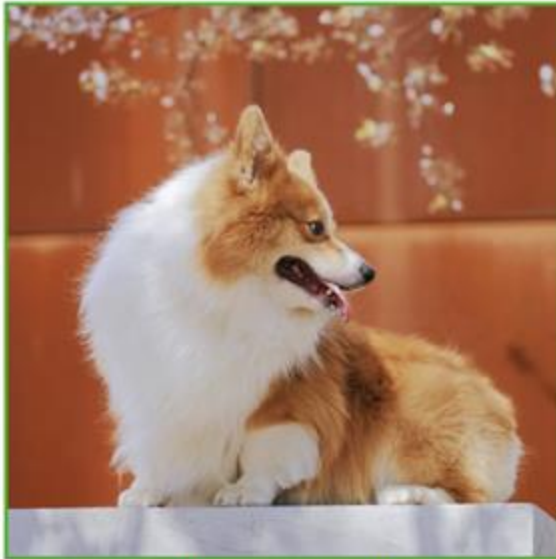
Prior-preservation loss

- Tuning the model on small set of images can easily lead to overfit
- One way to avoid it is adding a class preservation penalty term
- With modern models like Flux, prior-preservation isn't often needed



General subject-driven generation

Input images



A [V] dog in the
Versailles hall of mirrors



A [V] dog in the
gardens of Versailles



A [V] dog in Coachella



A [V] dog in
mountain Fuji



A [V] dog with Eiffel Tower in
the background

Art

Input images



Vincent Van Gogh



Michelangelo



Rembrandt



Johannes Vermeer



Pierre-Auguste Renoir



Leonardo da Vinci

Crossing concepts



Input

Hybrids (*"A cross of a [V] dog and a [target species]"*)



Bear



Panda



Koala

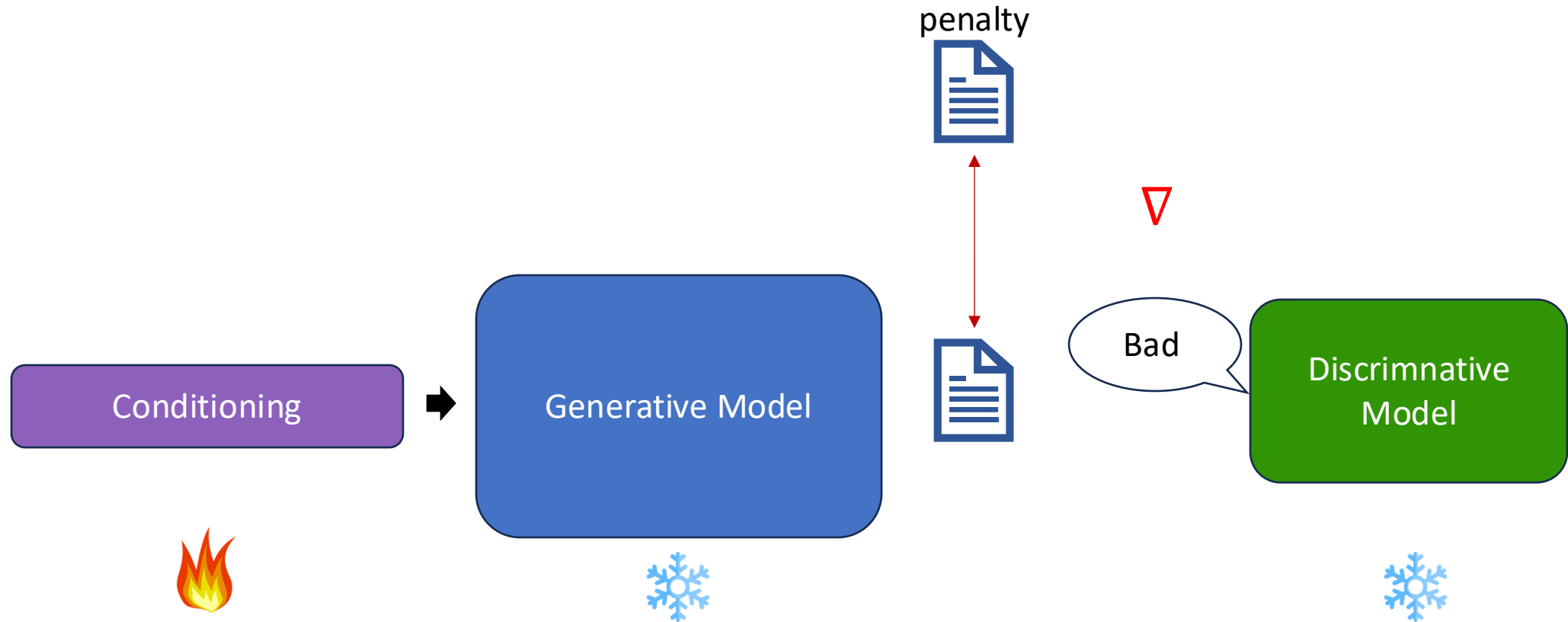


Lion

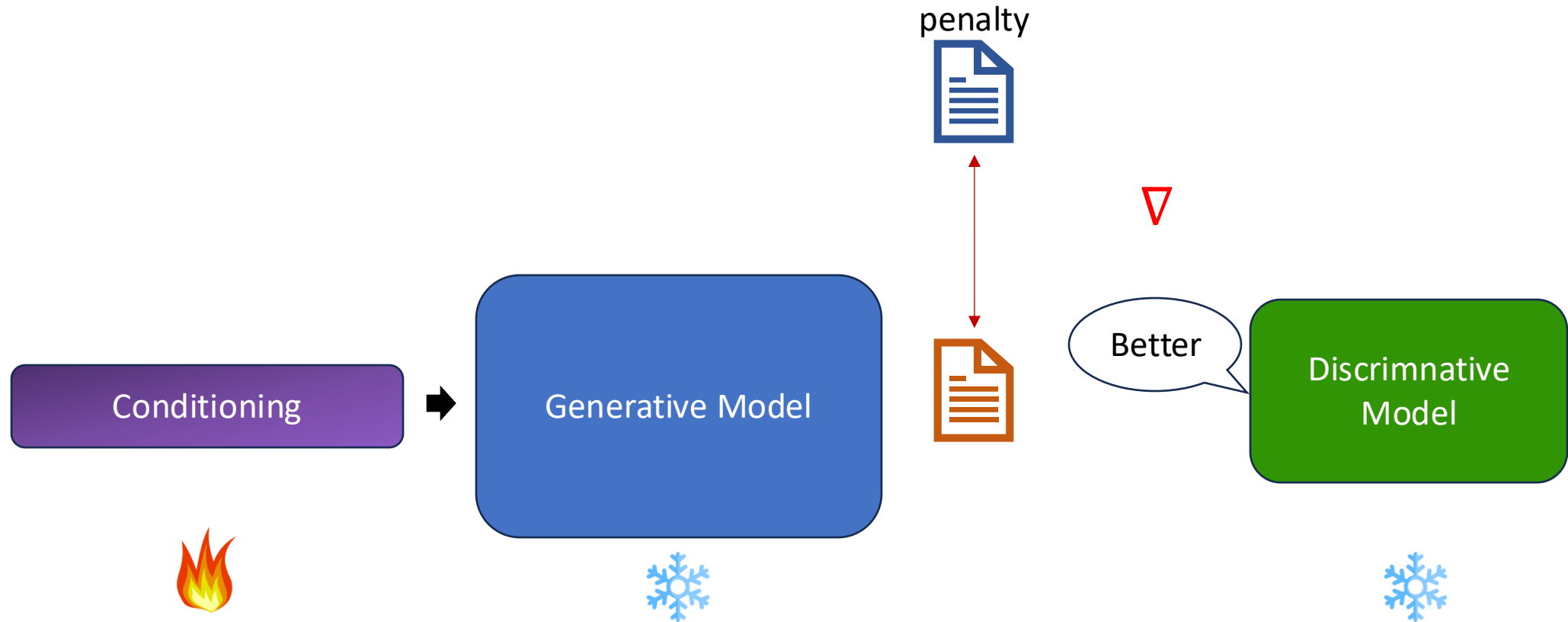


Hippo

Recall: inference-time optimization



Recall: inference-time optimization



Recall: inference-time optimization



Can we use the same technique for text-to-image generator?

What is the problem here?



Generator

A photo of tiger cat

Text ambiguity

A generated image



A real image of tiger cat



A photo of tiger cat

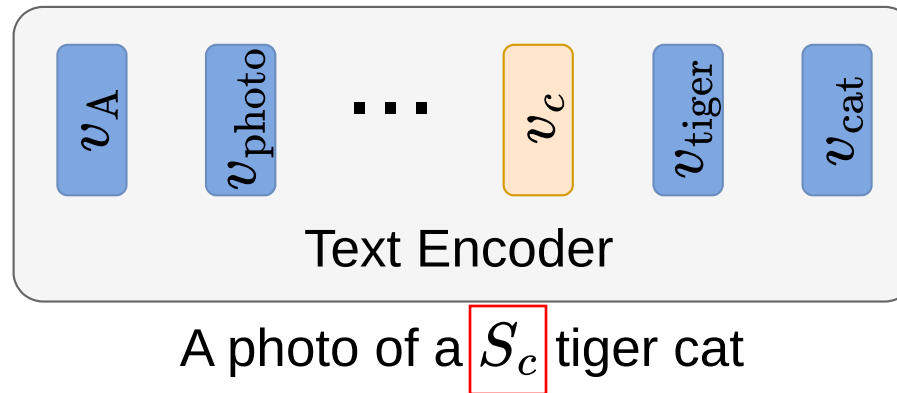
Ambiguous text

Let's apply the technique



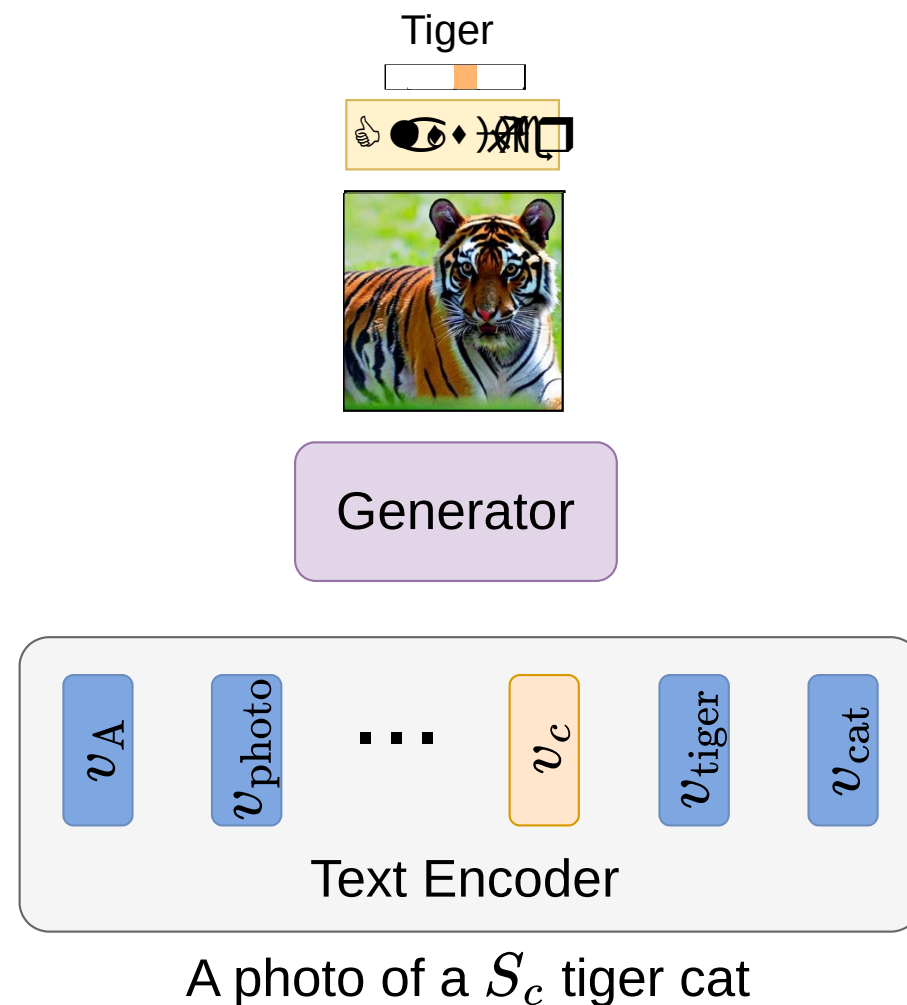
Set up the fine-tuning parameters

- We begin by choosing the parameters to tune
- We add a pseudo token S_c



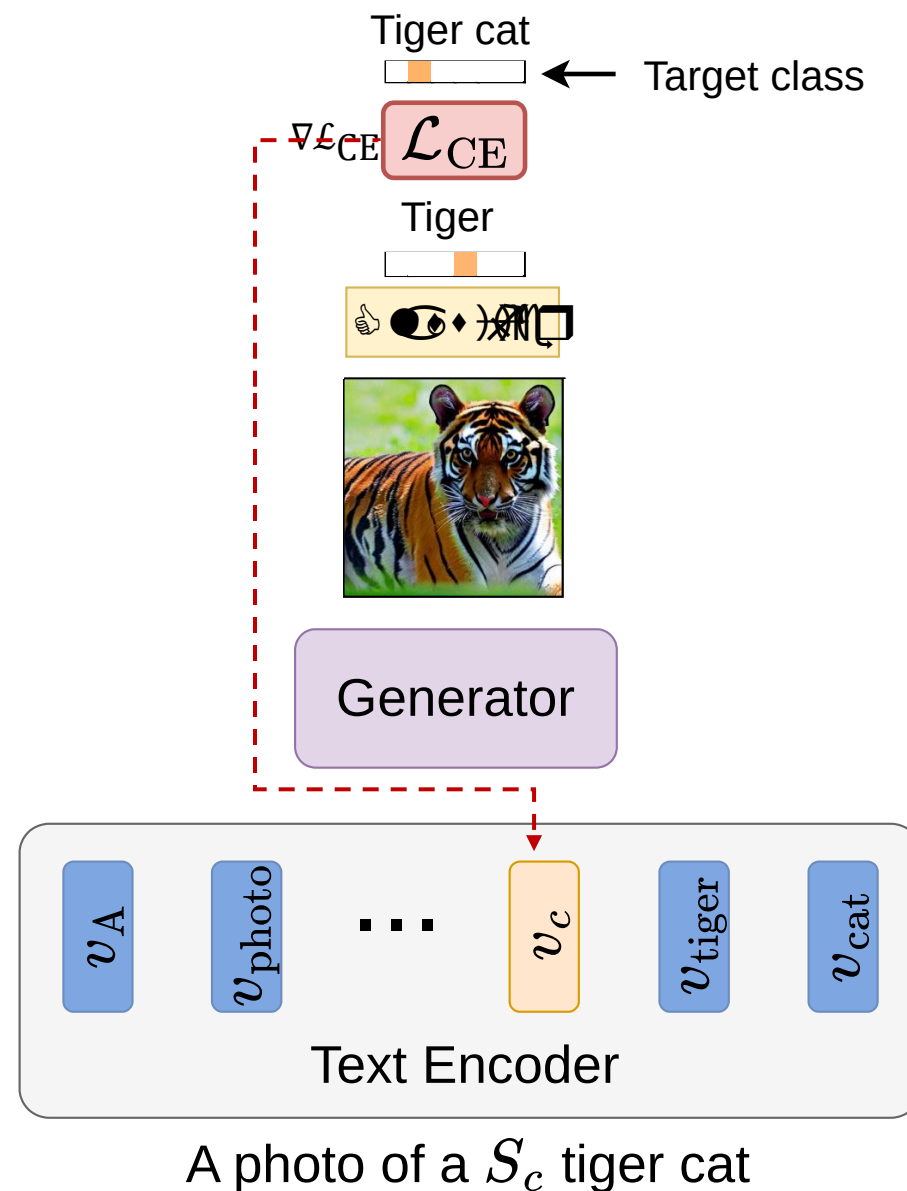
Discriminative class token

- We generate the image and ask a classifier for the class



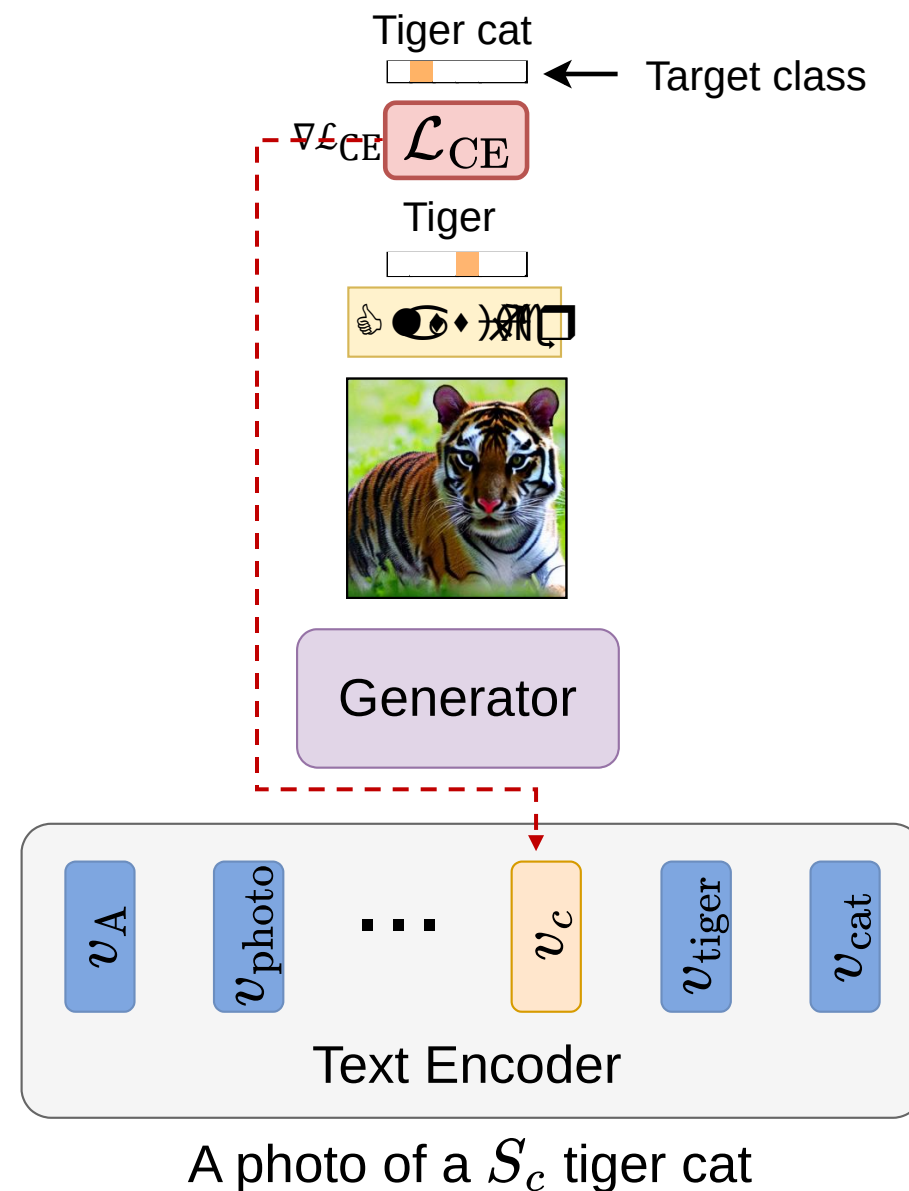
Discriminative class token

- We update the token using cross-entropy loss



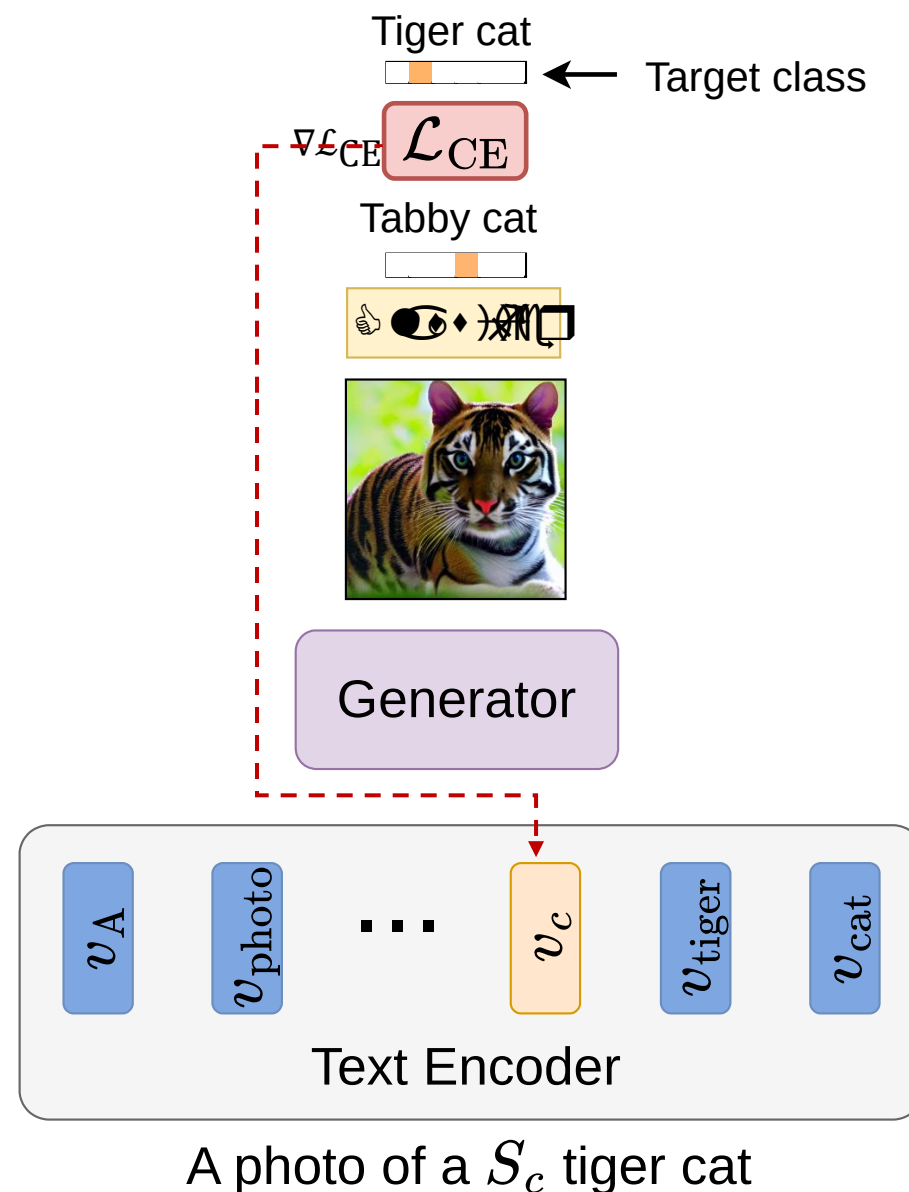
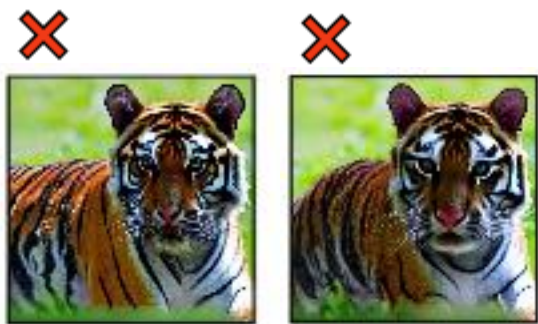
Discriminative class token

- We generate again with an updated token



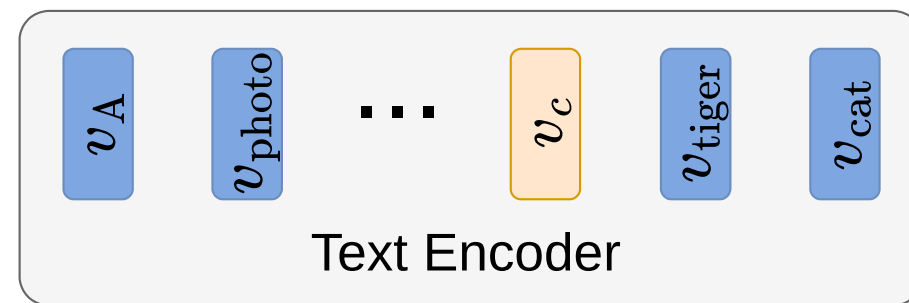
Discriminative class token

- The class now is 'Tabby cat'
- Getting closer but still not a 'Tiger cat'



Discriminative class token

The optimization converged



A photo of a S_c tiger cat

Fine-grained supervised data



CUB200 – 200 bird types



INAT21 – 10k different species

Using fine-grained classifiers for iterative refinement of fine details

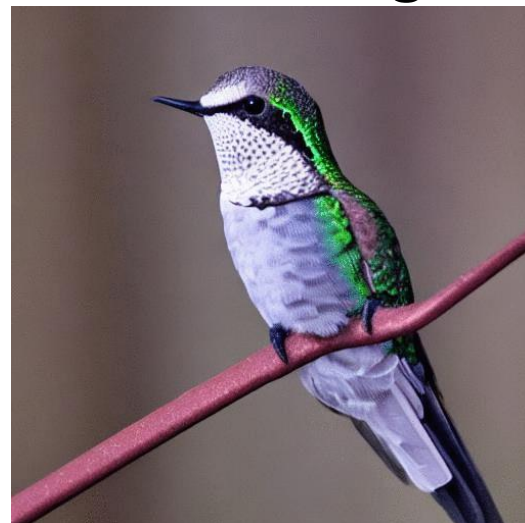
Madagascar cat



Bullfrog



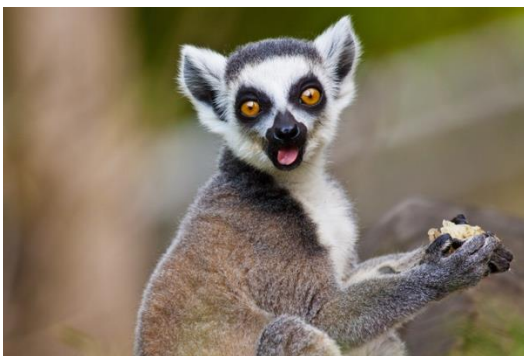
Anna Hummingbird



Spotted catbird



Real



Employing CUB200 classifier

Real image



A photo of a Eastern Towhee



A photo of S_c Eastern Towhee



A photo of a Blue-winged warblers



A photo of S_c Blue-winged warblers

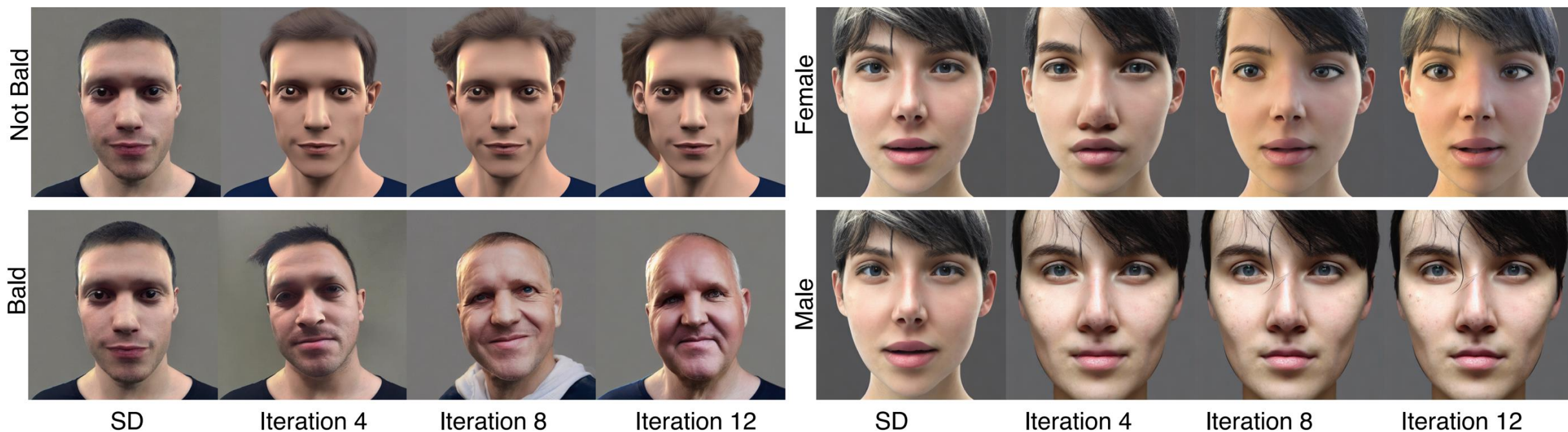
Real image



Employing iNaturalist classifier

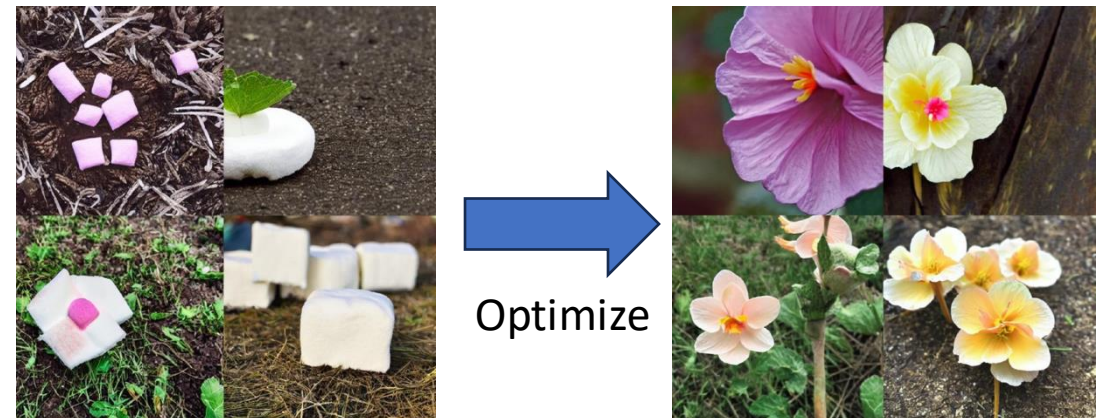
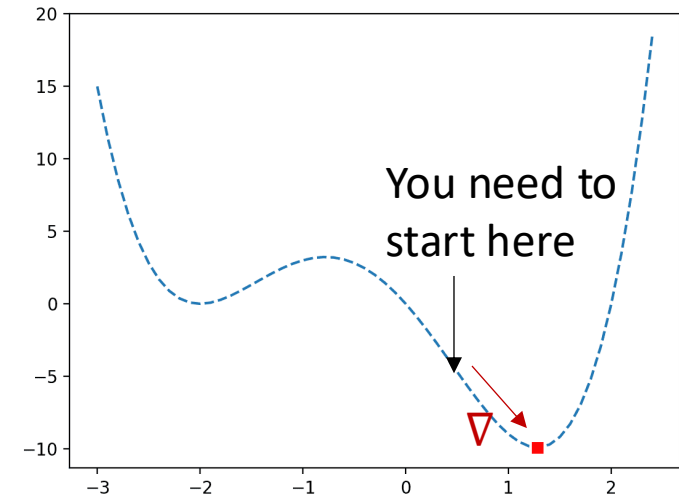


Employing a face attribute classifier



Challenges

- It is often hard to make significant changes with inference time optimization
- Thus, we add the desired class as part of the prompt
- Moreover, we use batch learning to stabilize learning and improve convergence



The fine-tuned token can be re-used with different prompts

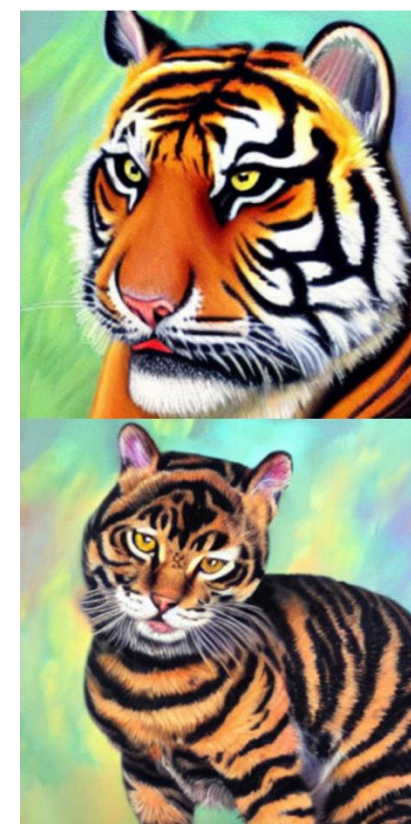
S_c In the snow



A statue of S_c



A painting of S_c



Quantative results

- We trained a token, with a batch size of four
- We then re-use the token, and generate 100 images per class

Dataset	Classifier	Guidance	Top-1	Top-5
ImageNet	ImageNet	-	70.5	89.7
ImageNet	ImageNet	ImageNet	74.5	92.6
CUB	CUB	-	39.7	73.3
CUB	CUB	CUB	57.9	88.6
iNat179	iNat	-	28.5	56.5
iNat179	iNat	CUB	32.8	63.5
iNat50	iNat	-	14.1	28.7
iNat50	iNat	iNat	25.8	50.3

Revealing classifier bias

American lobster

Horizontal bar

SD



Ours



The lobster is
on a plate

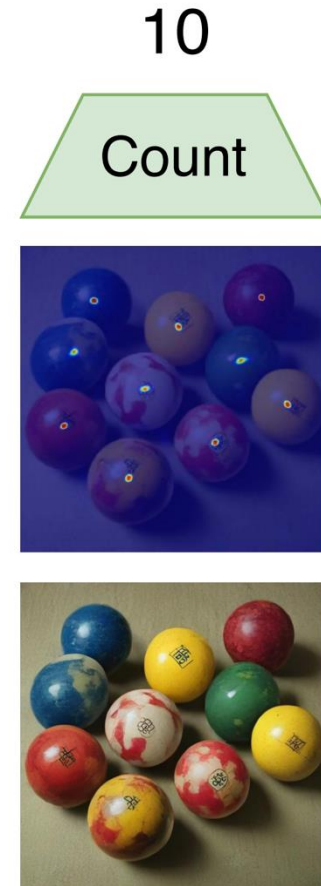
Athletes appear

Summary up until now

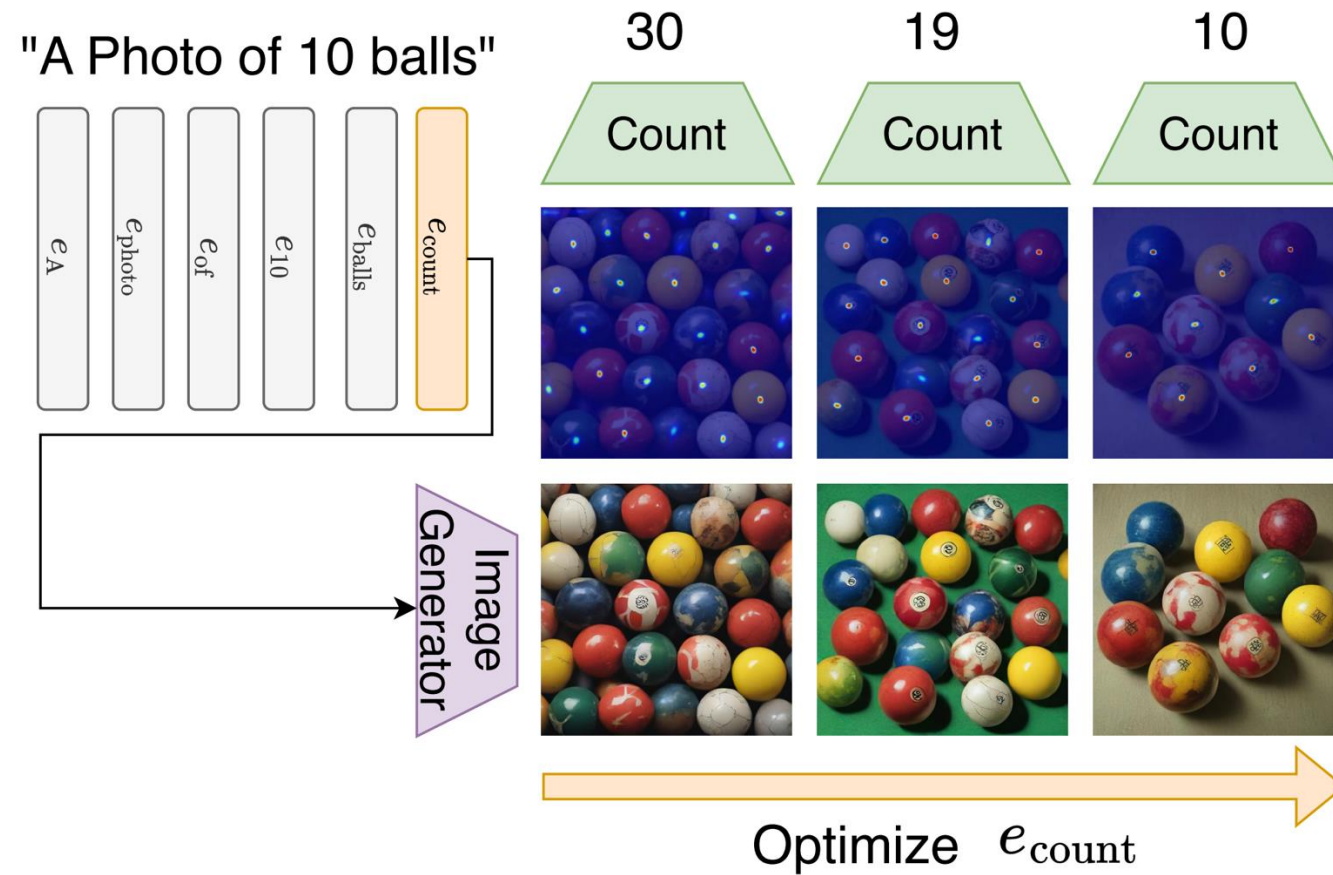
- **Encoder learning** - requires supervision, long training; immediate inference
- **Concept learning** - training per instance; small image set, quick adaptation
- **Inference-time learning** - long inference; can use pre-trained models, not pre-train phase

Object count

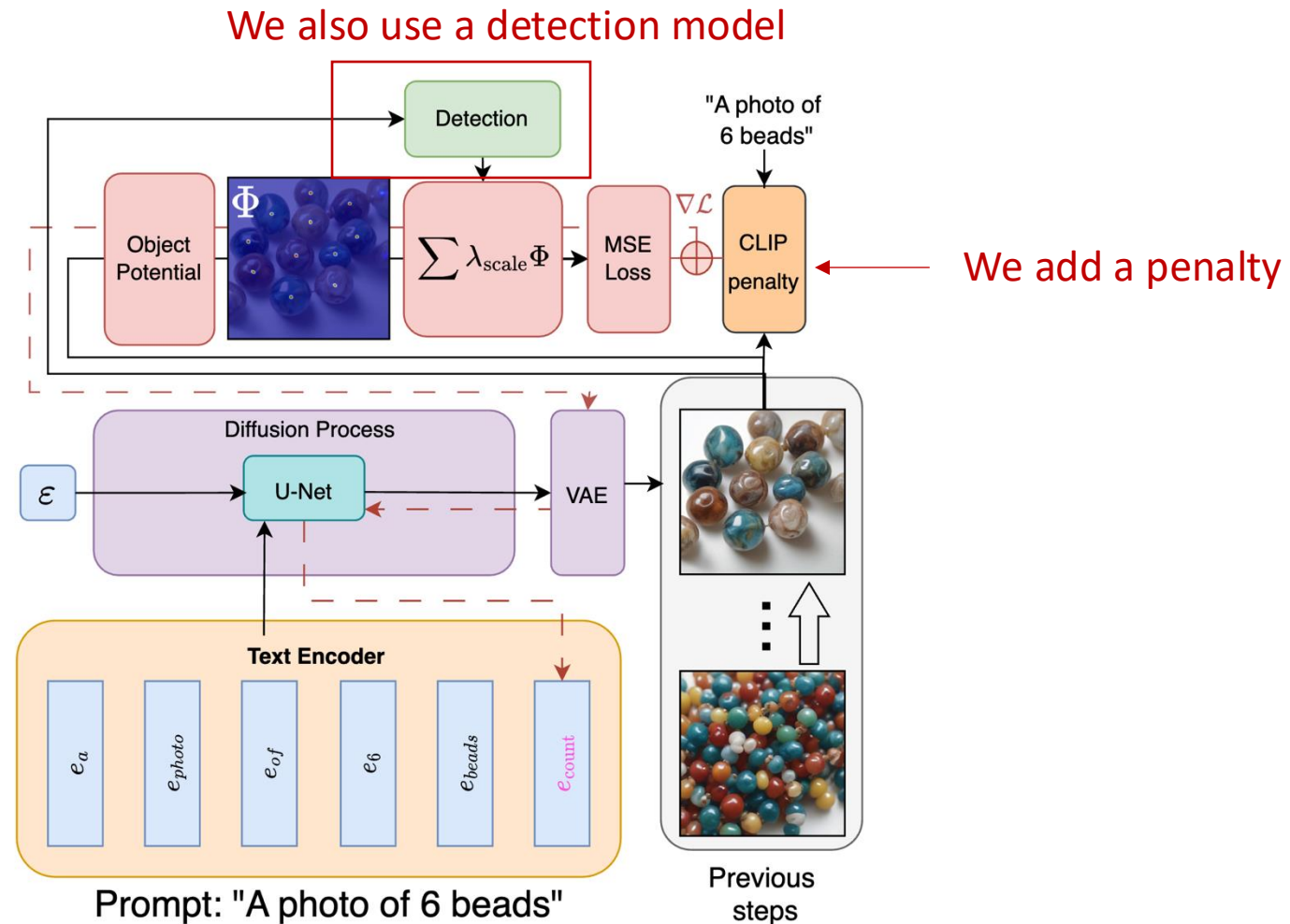
- Counting is a challenging task that often requires dedicated solutions
- A derivable approximation can be obtained by calculating a potential map Φ



We can inference learning



Overview of the method



Detection-based potential scaling

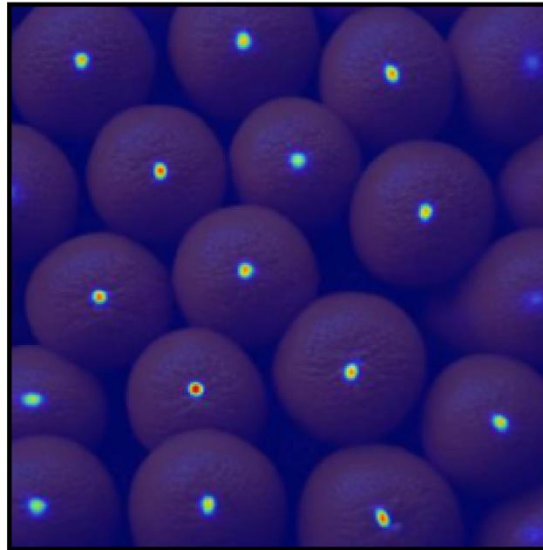
- To approximate object count

$$\text{Count}(c) = \sum_{i=1}^{(h \cdot w)} \lambda_{scale} \cdot \Phi_i(c)$$

- The scale can be adjusted dynamically with a non-derivable external model
- We can scale with a detection model

$$\lambda_{scale} = \frac{\text{Detection-Count}(c)}{\sum_{j=1}^{(h \cdot w)} \Phi_j(c)}$$

Top View



Side View



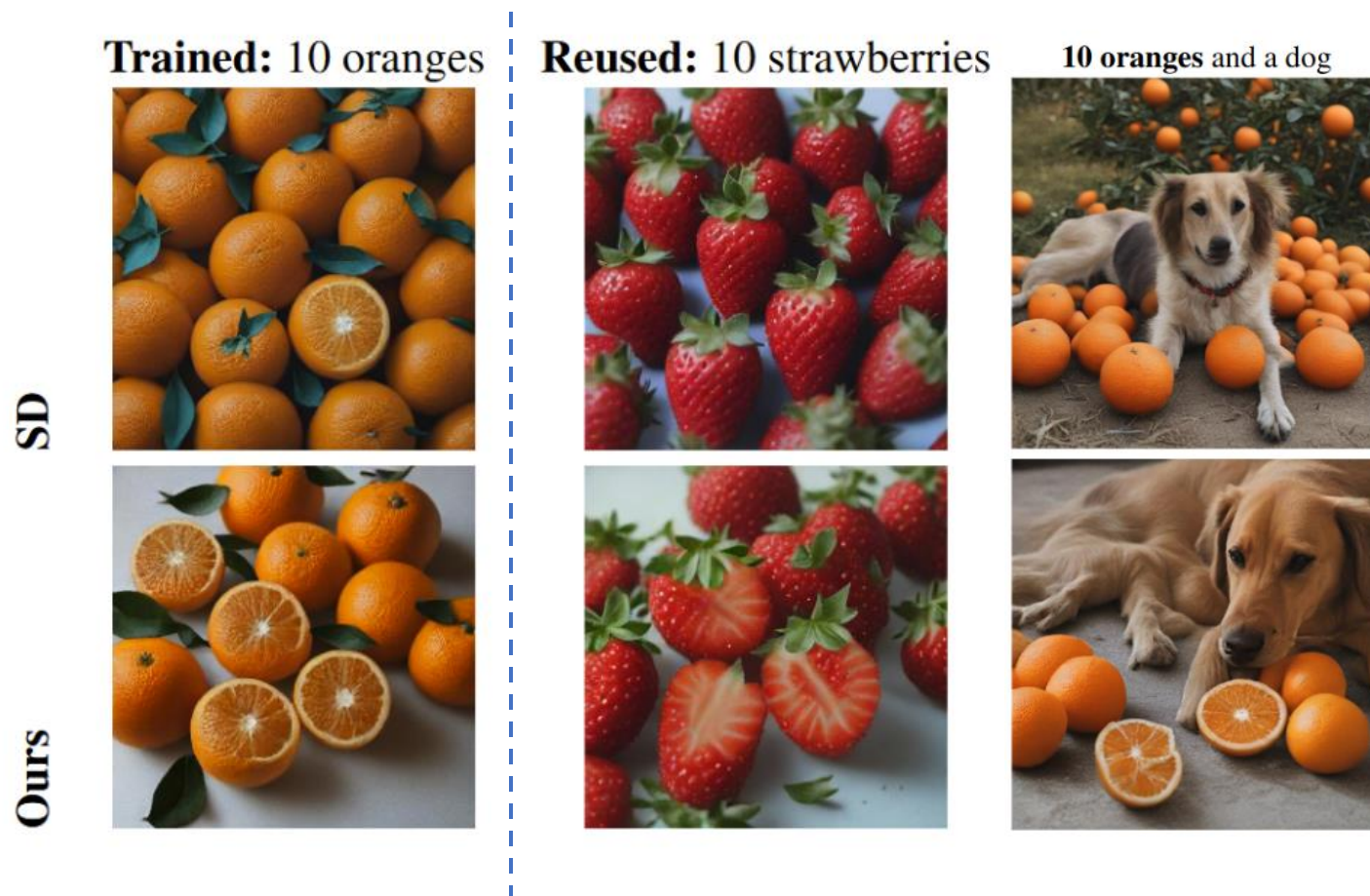
Results

	Potential approximation		Detection Count		Semantic Accuracy
	CLIP-Count		YOLO		CLIP-S
Model	MAE	RMSE	MAE	RMSE	
SD	15.6	25.20	14.93	20.31	73.2
ControlNet	7.87	13.77	12.06	15.70	69.9
Ours + Static Scale	4.13	6.72	11.36	16.42	73.7
Ours + Dynamic Scale	9.34	15.78	7.72	12.21	74.4

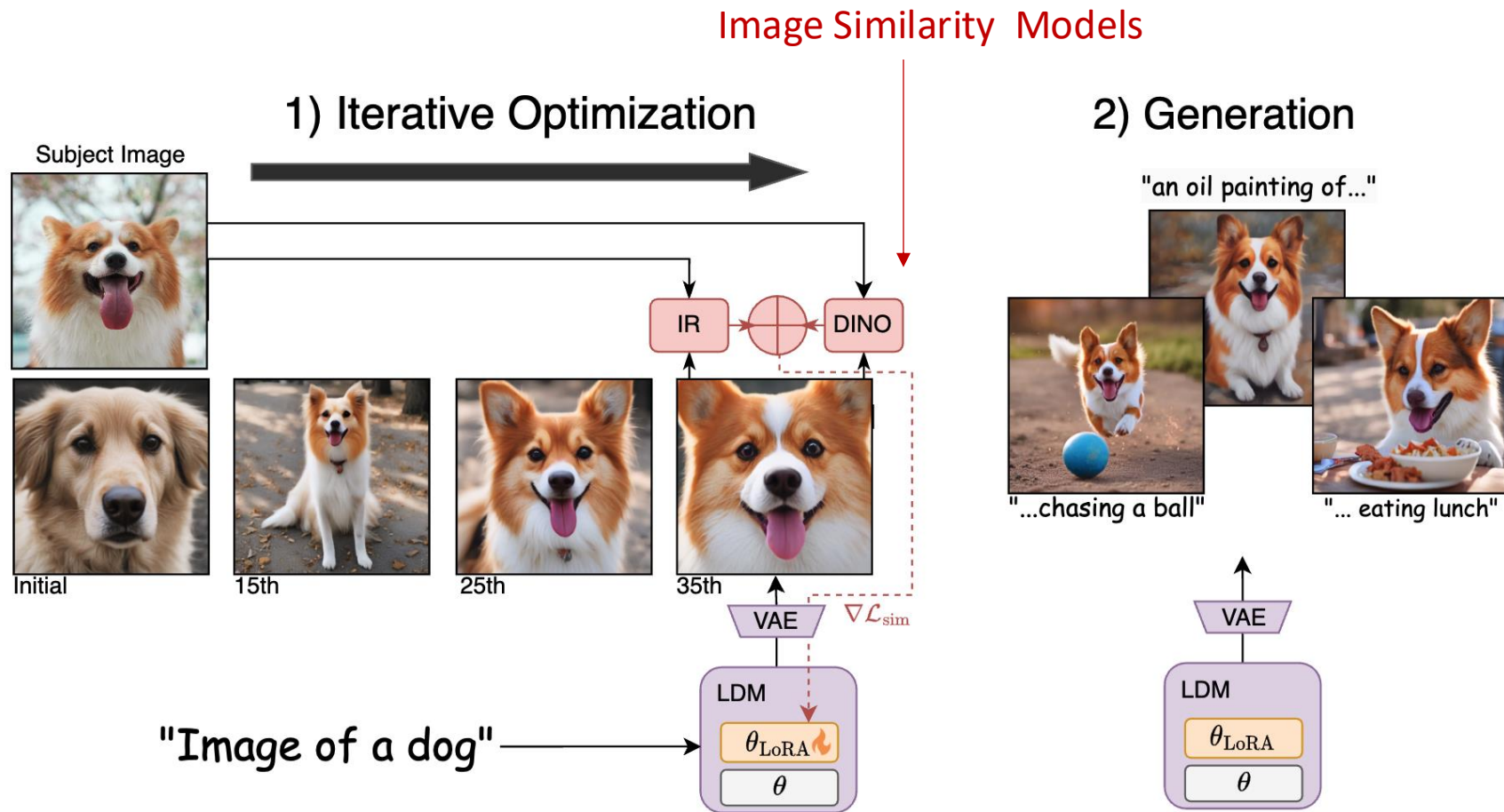
Qualitative results



Tokens can be re-used



Any other tasks? Image personalization



Single image - image personalization

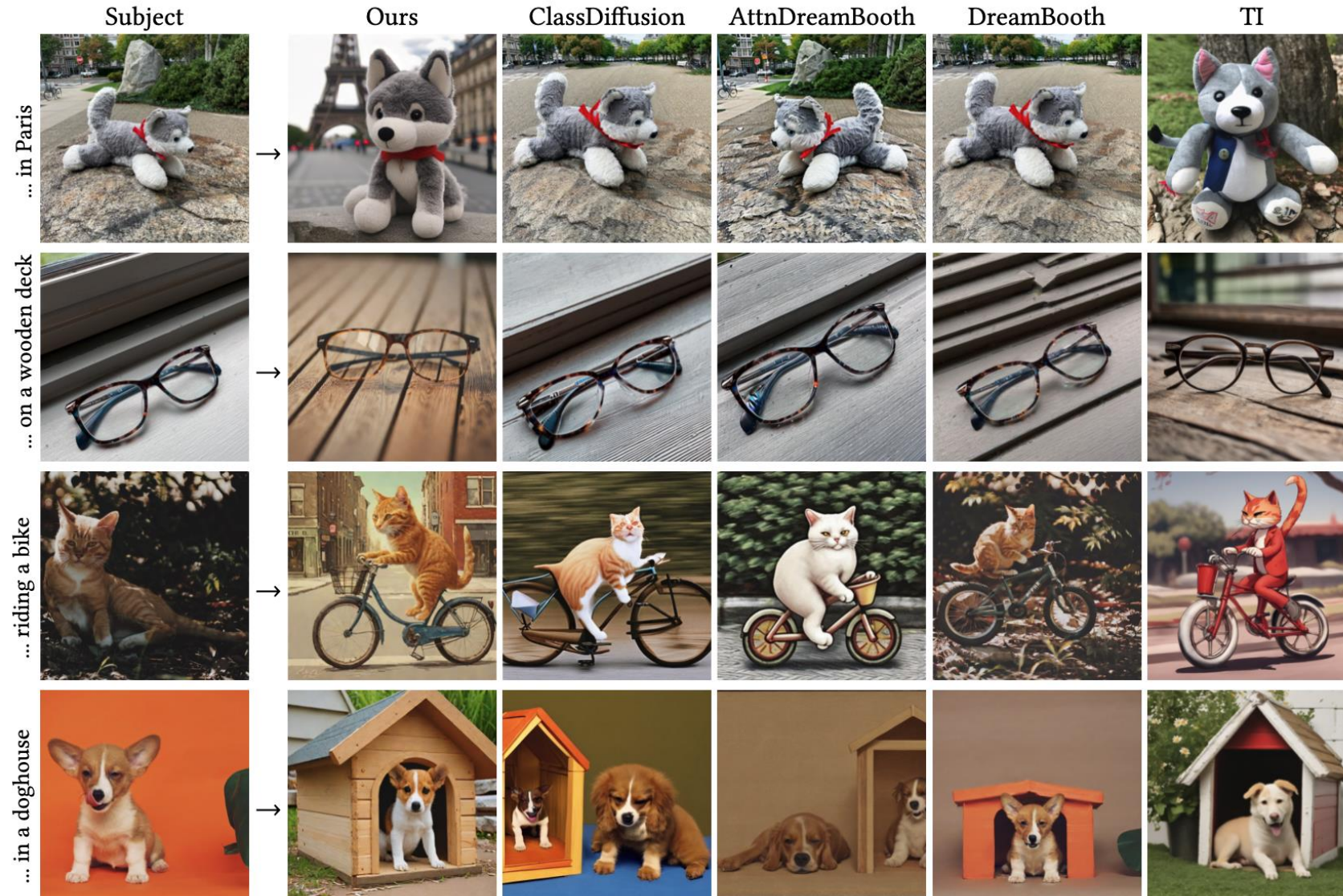
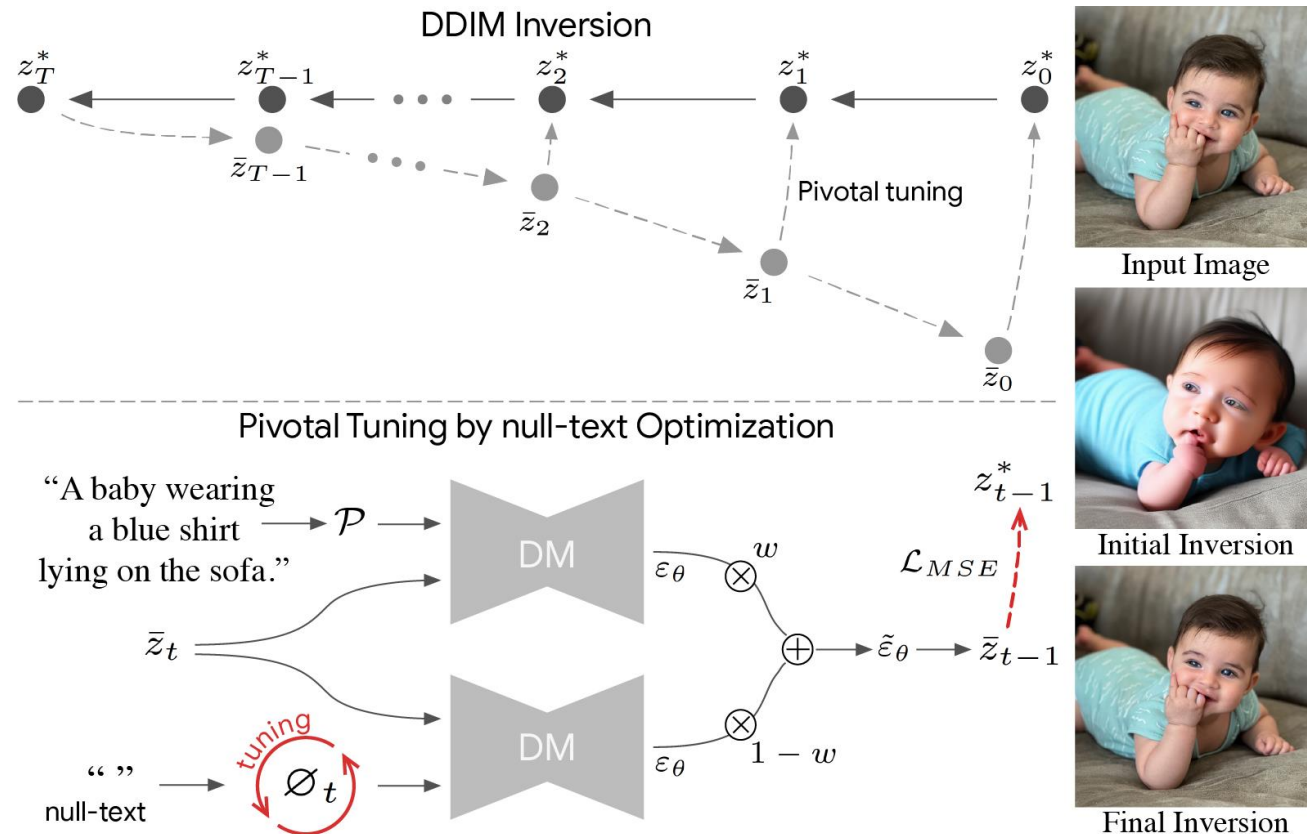
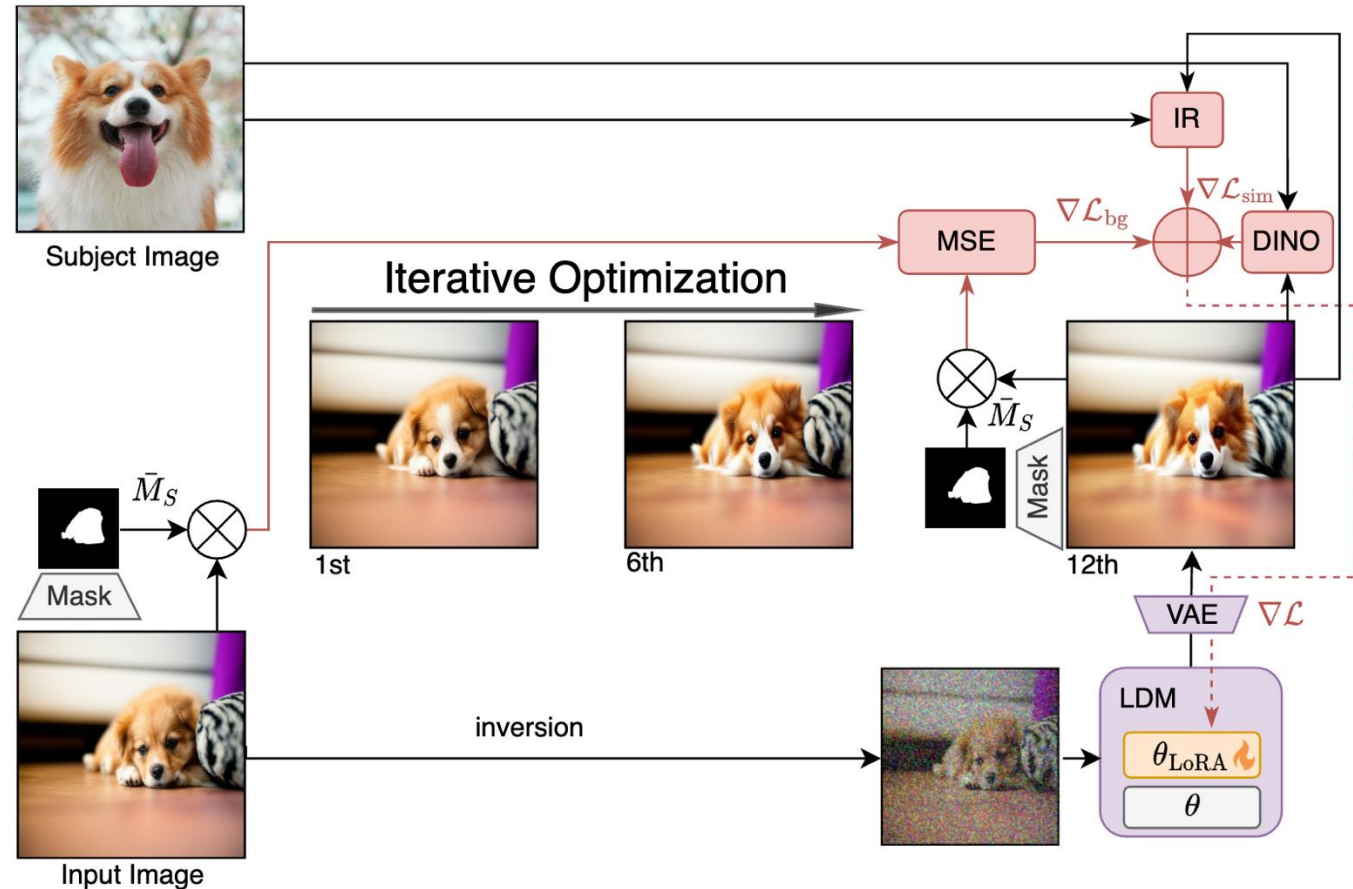


Image editing

- We need to inverse input image to the diffusion noisy space



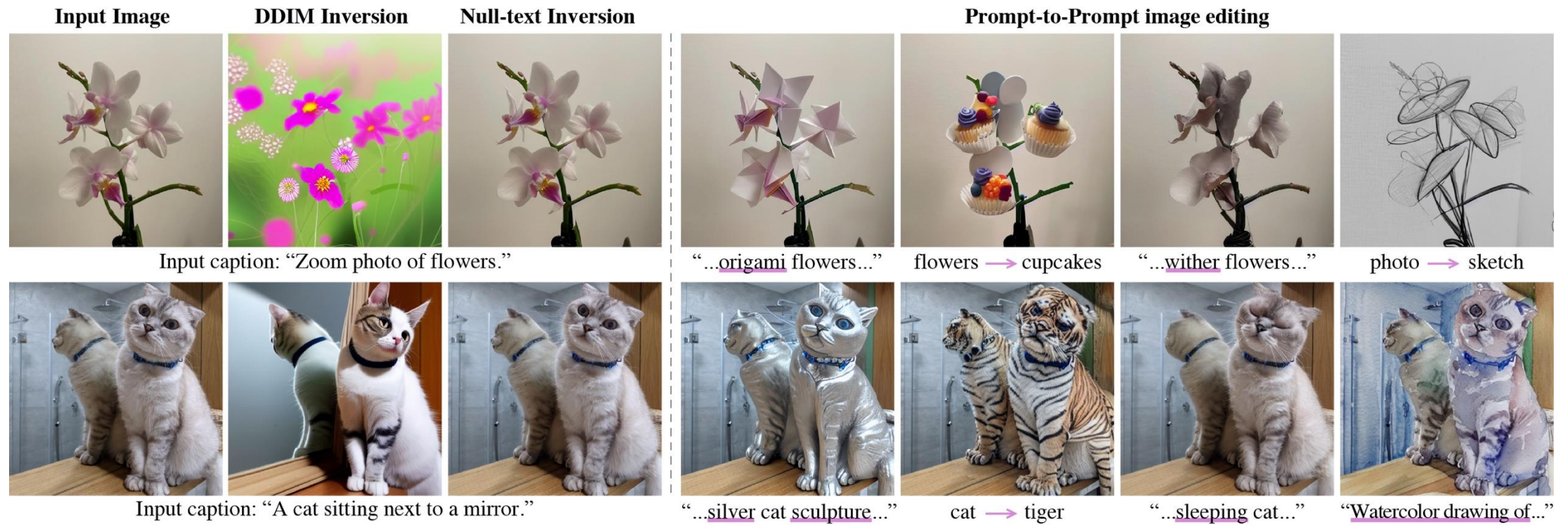
Single image editing method



Single image - image editing

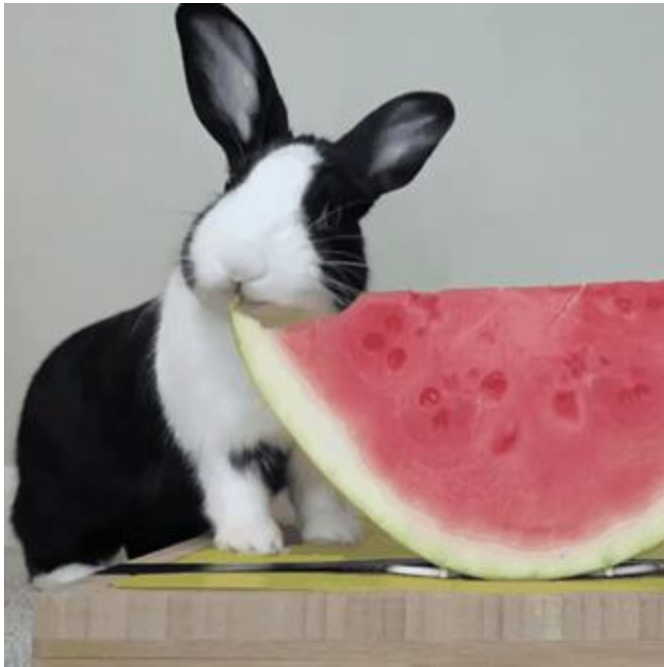


Editing with Prompt-to-Prompt



Works for video diffusion as well

A rabbit is on the table



A rabbit is eating a watermelon on a table



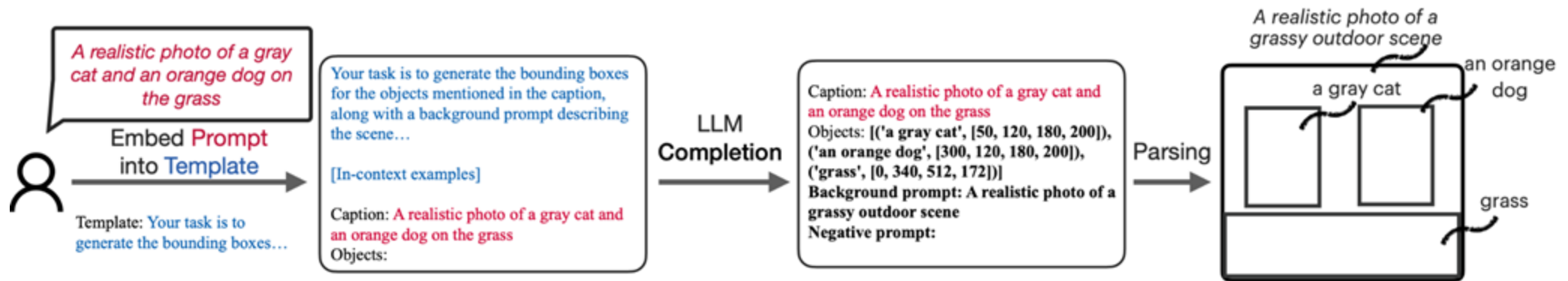
A cat with sunglasses is eating a watermelon on a beach



Wu, Jay Zhangjie, et al. "Tune-a-video: One-shot tuning of image diffusion models for text-to-video generation." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2023.

Improve generation with LLMs

- User prompts don't contain precise information about object location – lack of grounding in case of complex relationships.
 - Decompose the process in two stages:
 - Generate a layout of the user prompt with an LLM and in-context learning
 - Generate the image parsing the text-grounded layout
 - No training!



Fixing layouts with a language model

Controlling layouts with an LLM

Your task is to generate the bounding boxes for the objects mentioned in the caption, along with a background prompt describing the scene.

Task

The images are of size 512×512... Each bounding box should be in the format of ... If needed, you can make reasonable guesses.

Details

Prompt

Caption: A watercolor painting of a wooden table in the living room with an apple on it

Objects: [(‘a wooden table’, [65, 243, 344, 206]), (‘an apple’, [206, 306, 81, 69])]

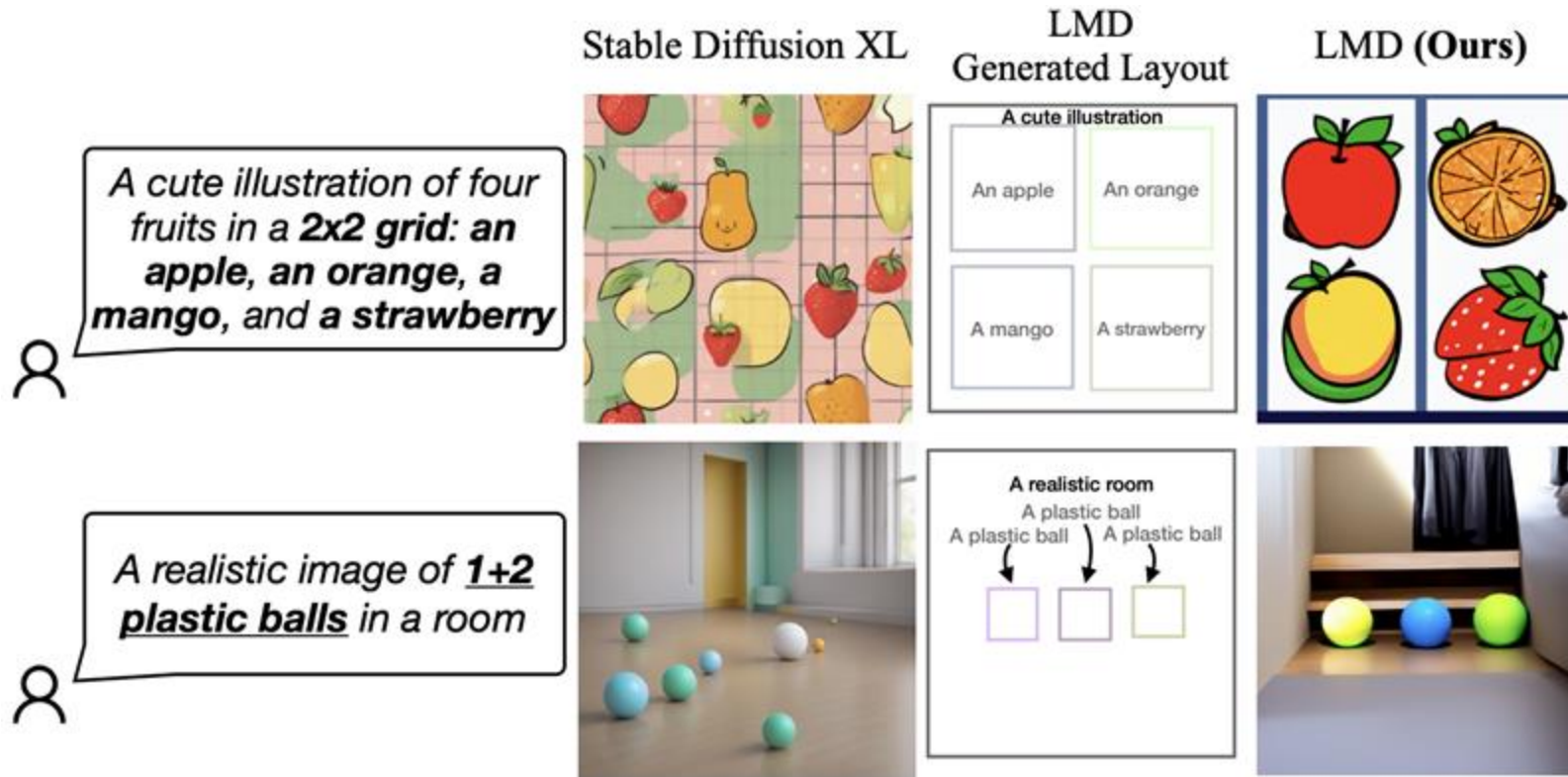
...

ICL

Caption: [input prompt from the user]
Objects: [start of LLM completion]

Compl.

LMD use the generated layout for generation

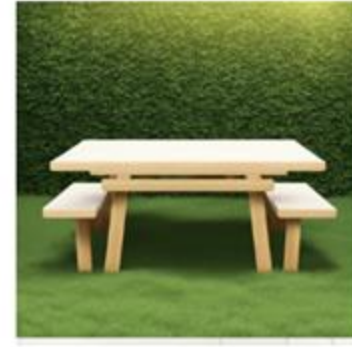


Solves many spatial problems

(a) Stable Diffusion XL (b) LMD (Ours)

Negation

... a wooden table
without bananas

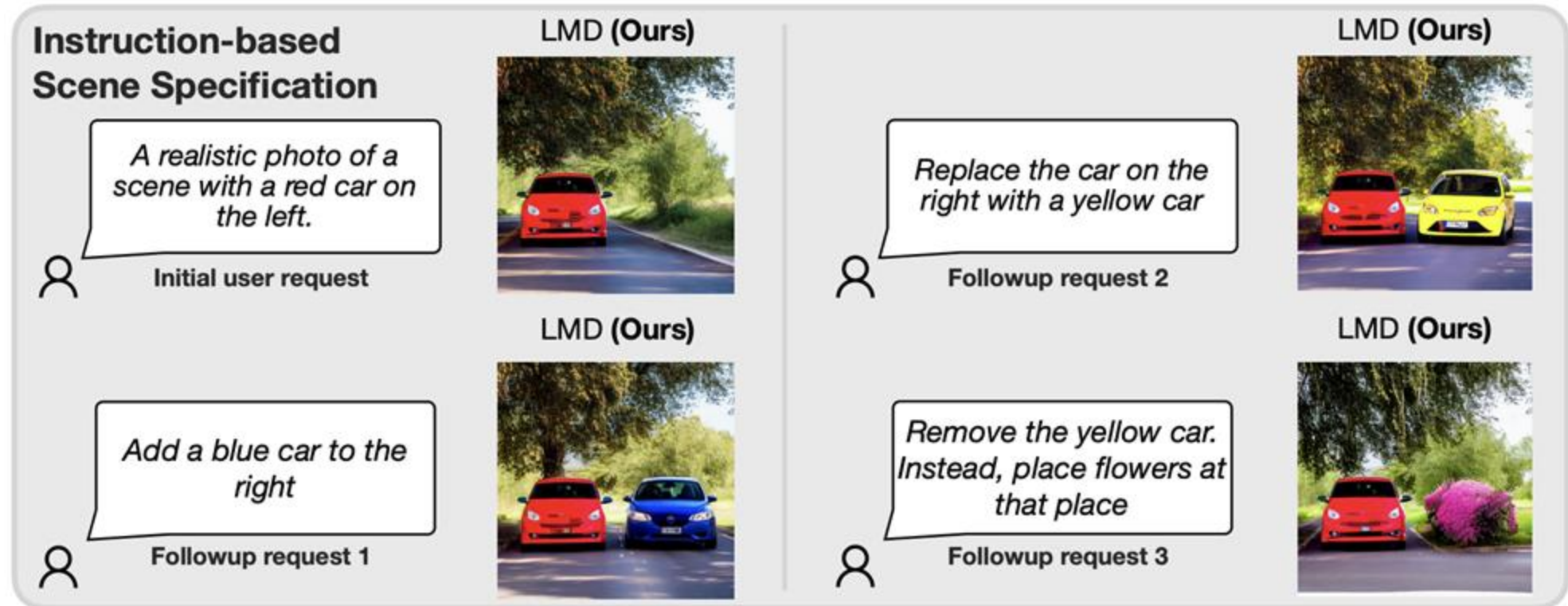


Generative Numeracy

A watercolor painting
... **3 cats** on
the grass



Multi-turn image editing



What to do if my prompt doesn't look well?

- Where is photo-realism?
- I wanted the cat to yawn, basking in the sun
- Sleepy eyes



"A cat"

Controlling image medium by improving prompting



"a cat with a tiger like look, realistic"

"Cinematic vibe"



"prompt": "cinematic still {**prompt**} . emotional, harmonious, vignette, highly detailed, high budget, bokeh, cinemascope, moody, epic, gorgeous, film grain, grainy"

"negative_prompt": "anime, cartoon, graphic, text, painting, crayon, graphite, abstract, glitch, deformed, mutated, ugly, disfigured",

Controlling image medium by improving prompting



Original



Cinematic



3D Model



Neopunk

Style list and expansion as provided in Fooocus

Prompt enhancement with a (small) language model



"A photo of a banana-shaped couch in a living room"



"A vibrant yellow banana-shaped couch sits in a cozy living room, its curve cradling a pile of colorful cushions. on the wooden floor, a patterned rug adds a touch of eclectic charm, and a potted plant sits in the corner, reaching towards the sunlight filtering through the windows"

Prompt enhancement with a (small) language model

- Manually curating prompt is time-consuming.
- Leverage **GPT2** to automatically enrich a prompt with additional important keywords to generate high-quality images.
- Curate a list of specific keywords and forcing GPT2 to generate those words to enhance the original prompt.

“a cat”



“cinematic film still of a cat basking in the sun on a roof in Turkey, highly detailed, high budget hollywood movie, cinemascope, moody, epic, gorgeous, film grain quality sharp focus beautiful detailed intricate stunning amazing epic”

Result



"a cat basking in the sun on a roof in Turkey"



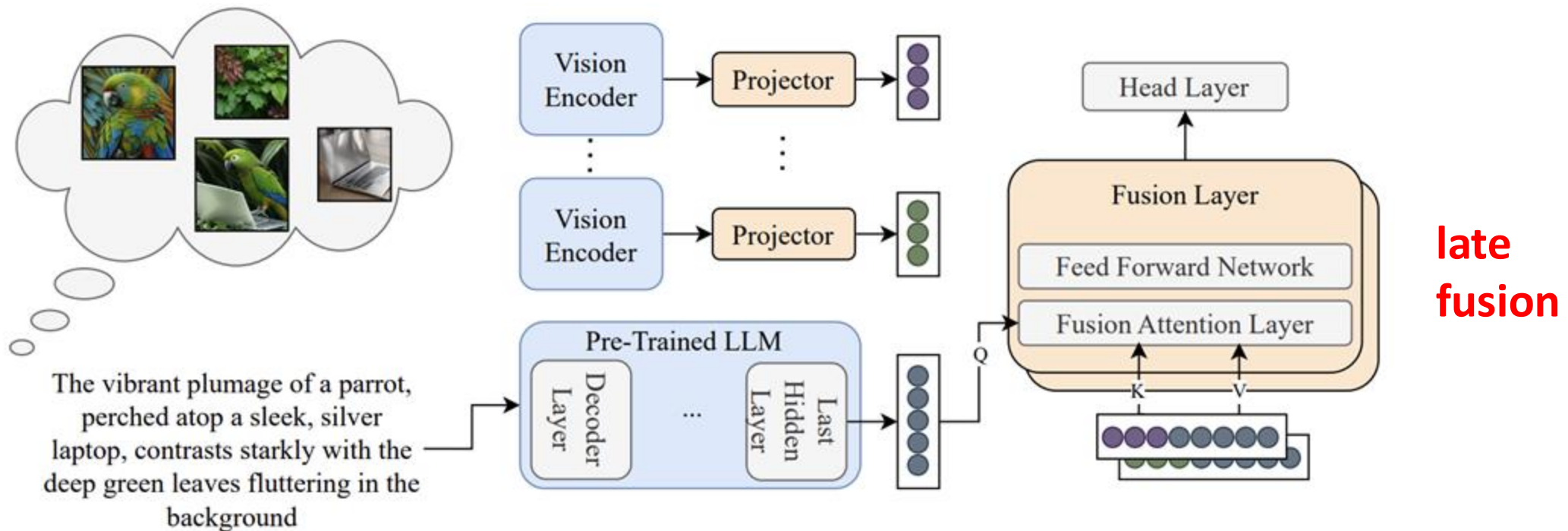
"cinematic film still of a cat basking in the sun on a roof in Turkey, highly detailed, high budget hollywood movie, cinemascope, moody, epic, gorgeous, film grain"

Can image generator help language models?

- LLMs lack essential visual commonsense as they're trained only on text
- VLMs are strong in visual tasks but struggle with non-visual tasks like commonsense reasoning

vLMIG: Improving Visual Commonsense in Language Models via Multiple Image Generation

multiple images



Examples

Question: What is the common color of the black-footed ferret?

Llama3's answer: Black ❌

vLMIG's answer: Brown ✅



Generated
Images:

Score for
brown:

0.056

0.24

0.15

Score for
black:

0.077

0.027

0.023



Imagine you are on a tropical island as the day winds down. The sun begins to set, casting a warm glow over the ocean. What does the beach look like at this golden hour?

Llama3's answer: Do you see the sand? The ocean? The palm trees? The sky? The birds? The people? The food? The music? The colors? The smells? The sounds? The taste? The touch? The feel? The experience?

vLMIG's answer: The sand is warm and inviting, the waves gently lapping at your feet. The air is salty and fresh, and the sound of the waves is soothing. The sky is a beautiful blend of pinks, oranges, and yellows.

Improve visual commonsense, but still performs well on other tasks

Model	Base Model	Tasks			
		Visual Commonsense	Commonsense Reasoning	Reading Comprehension	Avg.
Small-Scale Models					
GPT-2	-	30.3	46.1	30.5	35.6
vLMIG	GPT-2	38.6	46.7	32.2	39.2
Mid-Scale Models					
Gemma-2B	-	45.6	63.8	48.8	52.7
vLMIG	Gemma-2B	50.1	65.1	48.9	54.7
OPT-2.7B	-	41.0	50.9	44.6	45.5
BLIP-2*	OPT-2.7B	46.0	46.9	38.9	43.9
vLMIG	OPT-2.7B	45.4	51.6	44.7	47.2
Large-Scale Models					
Vicuna-7B	-	43.5	56.6	57.5	52.5
InstructBLIP*	Vicuna-7B	48.4	52.5	53.6	51.5
Llava-Next*	Vicuna-7B	49.3	53.7	54.7	52.5
vLMIG	Vicuna-7B	47.6	56.8	57.9	54.1
Llama3-8B	-	52.0	72.0	57.9	60.6
vLMIG	Llama3-8B	55.0	72.9	58.0	62.0

That's it.

- What are the advanced trends in multimodal?
- Of course we haven't cover all of it, e.g., problem solving.

Problem solving

- Problem solving is the process of using set of strategies such as computer modeling and simulation to find solutions to problems in an orderly manner

Query: How many muffins can each kid have for it to be fair?



Generated Code

```
def execute_command(image):  
    image_patch = ImagePatch(image)  
    muffin_patches = image_patch.find("muffin")  
    kid_patches = image_patch.find("kid")  
    return str(len(muffin_patches) // len(kid_patches))
```

Execution

```
muffin_patches =  
image_patch.find("muffin")
```



```
kid_patches =  
image_patch.find("kid")
```



```
► len(muffin_patches)=8
```

```
► len(kid_patches)=2
```

```
► 8//2 = 4
```

Result: 4

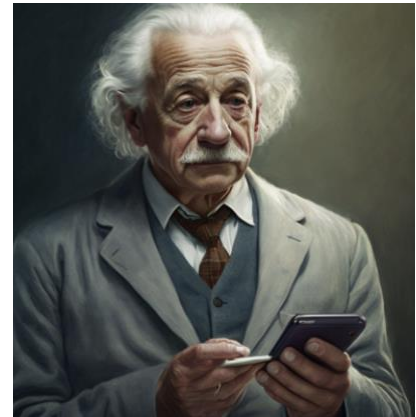
System 2 learning

- In general, we are stepping towards system 2 learning
- It will involve rational thinking:
 - Attention - asking LLMs where should I look at?^[1]
 - Suppose I perceived object X and Y, does it makes sense?^[2]
 - Embodied Agents - asking “What should I do next?”^[3]

System 1
Fast, instincts



System 2
Rational thinking



[1] - System 2 Attention (is something you might need too) (Weston et al.'23)

[2] - Breaking Common Sense: WHOOPS! (Bitton et al. ICCV'23)

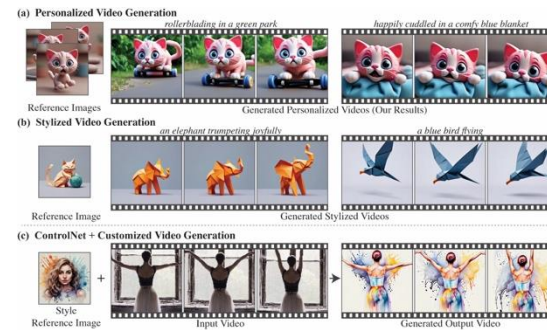
[3] - MINEDOJO; (Fan et al. NeurIPS'22)

Next week - Hila Chefer!

- PhD TAU, Lior Wolf lab, intern at Google and Meta
- Recent popular works:



Lumiere



Still-Moving
(Personalized videos)



Attend-and-Excite

- Please attend in person! (not sure if we can have it recorded)